

SLAM/SAP signaling regulates discrete $\gamma\delta$ T cell developmental checkpoints and shapes the innate-like $\gamma\delta$ TCR repertoire

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Somen K Mistri, Brianna M. Hilton, Katherine J. Horrigan, Emma S. Andretta, Remi Savard, Oliver Dienz, Kenneth J Hampel, Diana L. Gerrard, Joshua T. Rose, Nikoletta Sidiropoulos, Devdoot Majumdar, Jonathan E. Boyson 

Department of Surgery, Larner College of Medicine, University of Vermont, Burlington, Vermont 05405, USA •

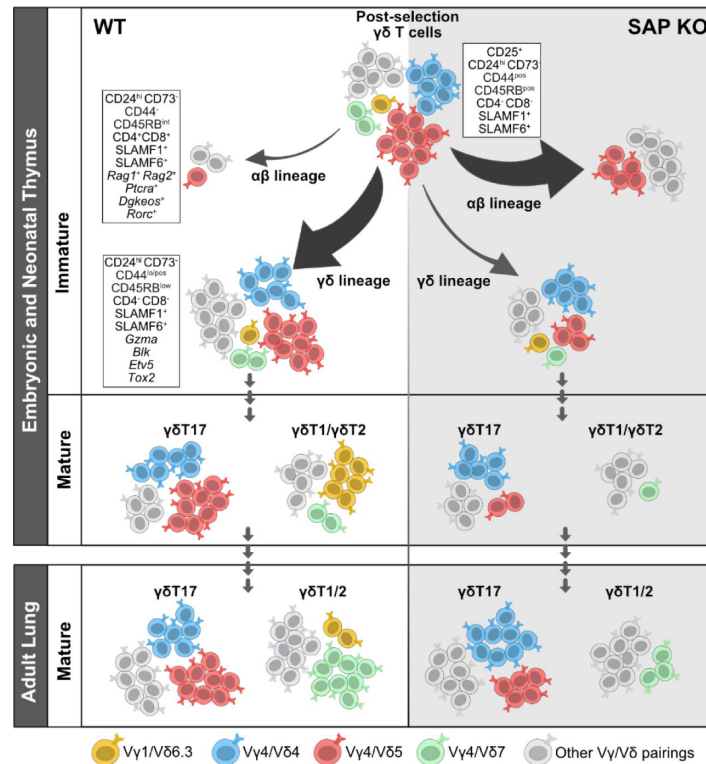
Department of Pathology and Laboratory Medicine, Larner College of Medicine, University of Vermont Medical Center, Burlington, Vermont 05405, USA

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Abstract

During thymic development, most $\gamma\delta$ T cells acquire innate-like characteristics that are critical for their function in tumor surveillance, infectious disease, and tissue repair. The mechanisms, however, that regulate $\gamma\delta$ T cell developmental programming remain unclear. Recently, we demonstrated that the SLAM-SAP signaling pathway regulates the development and function of multiple innate-like $\gamma\delta$ T cell subsets. Here, we used a single-cell proteogenomics approach to identify SAP-dependent developmental checkpoints and to define the SAP-dependent $\gamma\delta$ TCR repertoire. SAP deficiency resulted in both a significant loss of an immature $Gzma^+Blk^+Etv5^+Tox2^+$ $\gamma\delta$ T17 precursor population, and a significant increase in $Cd4^+Cd8^+Rorc^+Ptcra^+Rag1^+$ thymic $\gamma\delta$ T cells. SAP-dependent diversion of embryonic day 17 thymic $\gamma\delta$ T cell clonotypes into the $\alpha\beta$ T cell developmental pathway was associated with a decreased frequency of mature clonotypes in neonatal thymus, and an altered $\gamma\delta$ TCR repertoire in the periphery. Finally, we identify TRGV4/TRAV13-4(DV7)-expressing T cells as a novel, SAP-dependent V γ 4 $\gamma\delta$ T1 subset. Together, the data suggest that SAP-dependent $\gamma\delta/\alpha\beta$ T cell lineage commitment regulates $\gamma\delta$ T cell developmental programming and shapes the $\gamma\delta$ TCR repertoire.



- SAP deficiency results in diversion of $\gamma\delta$ T cells to the $\alpha\beta$ T cell lineage
- SAP deficiency results in a loss of immature $Gzma^{+}Bcl^{+}Etv5^{+}Tox2^{+}$ $\gamma\delta T17$ precursors
- Identification of $V\gamma4/V\delta7$ $\gamma\delta T1$ cells as a novel SAP-dependent innate-like T cell subset

eLife assessment

This **important** study highlights the role of SLAM-SAP signaling in shaping innate-like $\gamma\delta$ T cell subsets, providing **compelling** evidence for the importance of SLAM-SAP in immune system regulation, and the potential implications of the findings for tumor surveillance and infectious disease management. The work will be of broad interest to immunologists.

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Introduction

Like $\alpha\beta$ T cells, $\gamma\delta$ T cells can be broadly grouped into innate-like IFN- γ -producing $\gamma\delta T1$, IL-4-producing $\gamma\delta T2$, or IL-17-producing $\gamma\delta T17$ subsets that are programmed during thymic development^{1,2,3,4}, or inducible subsets that differentiate in the periphery⁵. The innate-like $\gamma\delta$ T cells are unconventional T cells that are highly enriched in mucosal tissues such as the skin, gut, and lung and that play critical roles in the host response to pathogens, in tumor surveillance, and in autoimmunity⁶. The mechanisms, however, that determine whether a developing $\gamma\delta$ T cell acquires an innate-like phenotype during thymic development remain unclear.

A significant body of data supports a model in which weak or strong TCR signal strength might contribute to the development of innate-like $\gamma\delta T17$ (weak signal) and $\gamma\delta T1$ (strong signal) subsets^{4,7,8,9,10}, and a strong TCR signal has been suggested to be necessary for the development of the IFN- γ and IL-4 producing $\gamma\delta$ NKT cell subset¹¹. However, there is also

evidence of a TCR-independent model in which pre-existing hard-wired transcriptional programs regulate the development of some innate-like $\gamma\delta$ T17 subsets¹², and accumulating data suggest that $\gamma\delta$ T17 programming is associated with the use of distinct signaling pathways^{13–15}.

SLAM family receptors comprise a family of nine receptors, SLAMF1 (SLAM; CD150), SLAMF2 (CD48), SLAMF3 (Ly9; CD229), SLAMF4 (2B4; CD244), SLAMF5 (CD84), SLAMF6 (Ly108; CD352), SLAMF7 (CRACC; CD319), SLAMF8 (BLAME), and SLAMF9 (SF2001; CD84H), that are expressed primarily on hematopoietic cells and which play diverse roles in immune development and function¹⁶. Most SLAM family receptors interact in a homophilic manner, the exception to this rule being SLAMF4 which serves as the ligand for SLAMF2. SLAM family receptor signals can be both activating or inhibitory, depending on the recruitment of the cytosolic signaling adapter proteins SAP and EAT-2, and inhibitory SHP-1, SHP-2, or SHIP phosphatases to immunoreceptor tyrosine switch motifs in the SLAM family receptor cytoplasmic domain¹⁶. Although SLAM/SAP signaling has long been known to play a role in innate-like NKT¹⁷ and $\gamma\delta$ NKT¹¹ cell development, recent findings from multiple groups demonstrating that this pathway is also involved in the development of innate-like $\gamma\delta$ T1, $\gamma\delta$ T17¹⁸ and innate-like MAIT cells^{19, 20} indicates that SLAM/SAP signaling plays a fundamental role in the development of a majority of the innate-like T cell lineages.

In this study, we set out to define the stage of thymic development during which SAP mediates its effect on $\gamma\delta$ T cell developmental programming and to define SAP-dependent changes in the $\gamma\delta$ T cells transcriptional program during their development. Finally, since SAP-dependent innate-like T cell subsets are often characterized by highly restricted TCRs (e.g., $V\gamma 1^+/V\delta 6.3^+$ $\gamma\delta$ NKT²¹), we wished to identify whether specific TCR clonotypes were also associated with SAP-dependent $\gamma\delta$ T1 and $\gamma\delta$ T17 T cells. Therefore, we utilized a single-cell proteogenomics approach with V(D)J profiling that allowed us to simultaneously compare cell surface protein expression, gene expression, and TCR clonotypes between B6 and B6.SAP^{-/-} $\gamma\delta$ T cells from embryonic, neonatal, and adult thymus, as well as from peripheral lung $\gamma\delta$ T cells. Collectively, these data provide a comprehensive map of the SAP-dependent $\gamma\delta$ T cell subsets and their developmental checkpoints in perinatal and adult thymus.

Results

SLAM family receptor expression marks transcriptionally distinct developmental stages among E17 $\gamma\delta$ T cells

$\gamma\delta$ T17 developmental programming is thought to occur primarily during embryonic/neonatal thymic development^{2, 12, 22}, and is tightly linked to the expression of discrete TCR gamma and delta pairings²³. To identify the specific developmental stages of thymic development during which SLAM/SAP signaling pathway exerts its effects, we employed a single-cell RNAseq approach coupled with feature barcoding (i.e., CITEseq²⁴) and V(D)J profiling to compare embryonic day 17 (E17) B6 and B6.SAP^{-/-} thymic $\gamma\delta$ T cells (**Fig. 1A**). FACS-sorted thymic $\gamma\delta$ T cells (**Suppl. Fig. 1**) from individual embryos (n = 4 B6 and 4 B6.SAP^{-/-} mice) were identified using hashtags, and $V\gamma 1^+$ and $V\gamma 4^+$ gd T cells were identified using barcoded antibodies (ADTs). To help define the developmental trajectories of the E17 gd T cells, we also stained cells with barcoded antibodies specific for CD24, CD73, CD44, and CD45RB which define distinct developmental pathways of $\gamma\delta$ T17 and $\gamma\delta$ T1 cells in the thymus^{4, 25, 26}, as well as for SLAMF1 and SLAMF6. After data filtering and QC, we verified that clusters were evenly distributed among the biological replicates (**Suppl. Fig. 1**). The resulting dataset represented a total of 5278 thymic $\gamma\delta$ T cells from B6 (n = 2597) and B6.SAP^{-/-} (n = 2681) $\gamma\delta$ T cells. Data were visualized using uniform manifold approximation and projection (UMAP) for dimensionality reduction and a graph-based clustering approach²⁷.

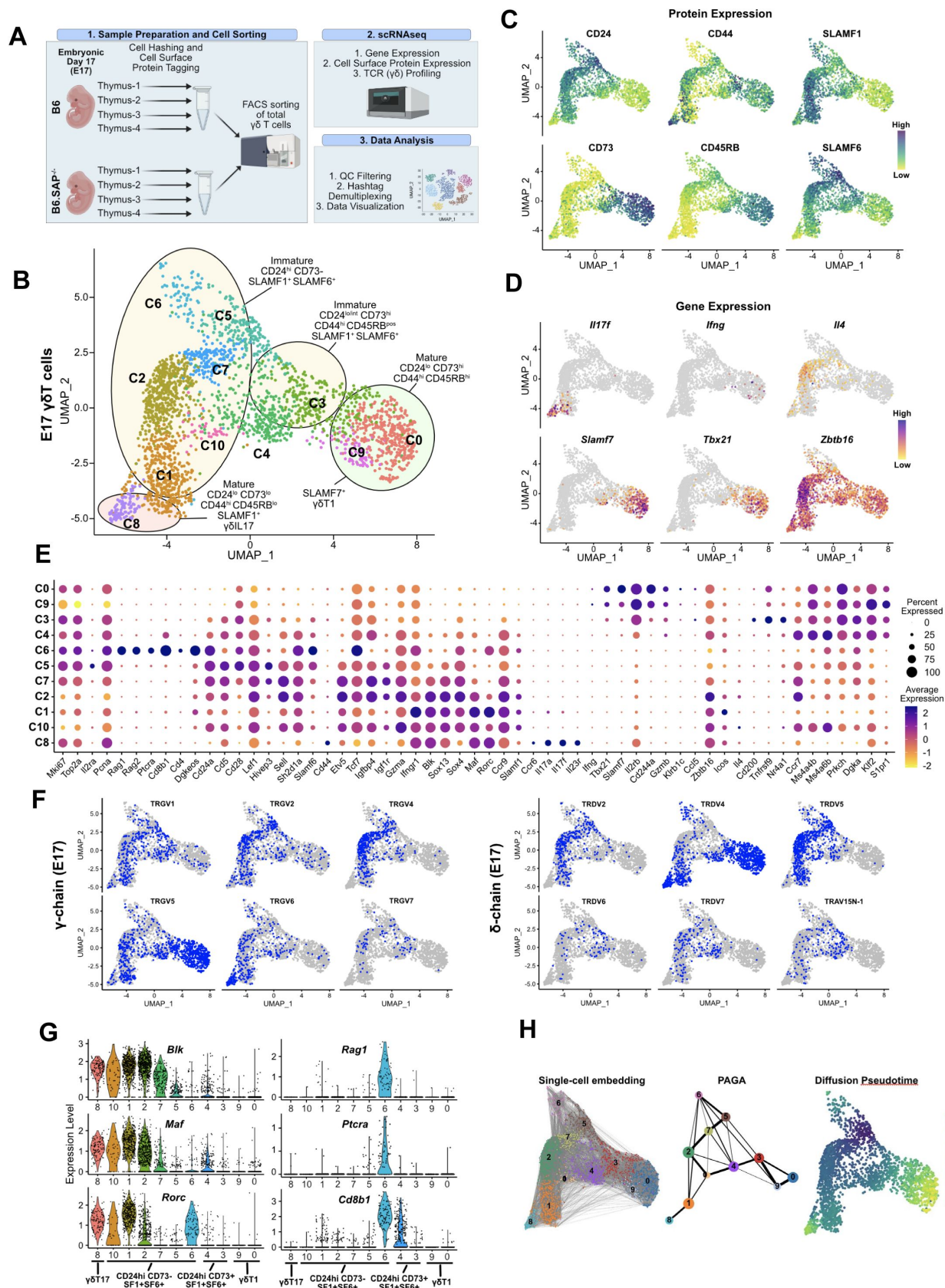


Figure 1.

SLAM family receptor expression marks transcriptionally distinct developmental stages among E17 $\gamma\delta$ T cells.

(A) Schematic workflow depicting the methodology for single-cell RNA sequencing (scRNAseq) library preparation and subsequent data analysis pipeline employed in this study. (B) Uniform Manifold Approximation and Projection (UMAP) visualization displaying 11 distinct clusters of E17 B6 thymic $\gamma\delta$ T cells, $n = 4$ individual mice. Clusters are annotated based on comprehensive protein and gene expression data. (C) Feature plots illustrating the cell surface protein expression profiles of CD24, CD73, CD44, CD45RB, SLAMF1, and SLAMF6 on B6 E17 thymic $\gamma\delta$ T cells. Each data point represents a cell, color-coded to indicate varying protein expression levels (high: dark blue, low: yellow). (D) Feature plot illustrating the gene expression profiles of signature genes among individual B6 E17 thymic $\gamma\delta$ T cells. Each data point represents a cell, color-coded based on gene expression levels (high: purple, low: yellow). (E) Dot plot demonstrating the scaled expression levels of selected genes in E17 B6 thymic $\gamma\delta$ T cells. Normalized expression levels are depicted using a color scale ranging from low (yellow) to high (purple). Dot size corresponds to the fraction of cells within each cluster expressing the specific marker. (F) UMAP representation of E17 B6 thymic $\gamma\delta$ T cells indicating the expression of selected TRGV (TCR γ ; left) and TRDV (TCR δ ; right) chain V-segment usage (in blue) across individual cells. (G) Violin plots illustrating the expression patterns of selected genes among E17 B6 thymic $\gamma\delta$ T cell clusters. (H) Visualization of single-cell trajectories using PAGA (partition-based graph abstraction) with single-cell embedding (left) showing connectivity between individual nodes (middle). Weighted edges represent statistical measures of interconnectivity. The diffusion pseudotime plot (right) delineates inferred pseudotime progression of cells along developmental trajectories using cluster 5 (C5) as the root, highlighting their developmental order (from purple to yellow).

First, we annotated the B6 dataset using a combination of cell surface protein expression, gene expression, and TCR clonotype sequence. This analysis revealed 11 B6 E17 $\gamma\delta$ T cell clusters, mostly distributed between two major branches (Fig. 1B). We observed the presence of CD24^{low}73^{low}44^{high}45RB^{low}SLAMF1^{pos} (c8) and CD24^{low}73^{high}44^{high}45RB^{high}SLAMF6^{low/pos} (c0, c9) clusters, consistent with mature $\gamma\delta$ T17 phenotype and $\gamma\delta$ T1 phenotypes, respectively (Fig. 1C, Suppl. Fig. 1). In support of this annotation, the c8 cluster was enriched in *Il17a*, *Il17f*, *Il23r*, *Rorc*, and TRGV6/TRDV4 (IMGT nomenclature; encoding Vy6/V δ 1) transcripts with limited CDR3 diversity, while the c0/c9 cluster was enriched in *Ifng*, *Tbx21*, *Klrd1*, *Il2rb*, *Xcl1*, and TRGV5/TRDV4 (encoding Vy5/V δ 1) transcripts also with little to no diversity (Figs. 1D – F, Suppl. Fig. 2, Suppl. Tables 1 and 2). Consistent with previous observations¹⁸, we noted that the c0/c9 cluster was highly enriched in *Cd244a* (*Slamf4*), and was also enriched in transcripts from another SLAM family member, *Slamf7* (Figs. 1D – E). Flow cytometric analysis confirmed that SLAMF7 was expressed primarily on mature $\gamma\delta$ T cells and that its expression was mostly confined to $\gamma\delta$ T cells expressing the canonical CD44⁺CD45RB⁺ markers of $\gamma\delta$ T1 cells (Suppl. Fig. 1).

We identified a small CD24^{low/int}73^{high}44^{high}45RB^{pos} c3 cluster characterized by enrichment in *Cd200* (whose expression is upregulated upon TCR ligation³¹), *Il2rb*, *Nr4a1*, *Cd244a*, *Xcl1*, and *Tnfrsf9* (encoding 4-1bb receptor). A high frequency of invariant TRGV5/TRDV4 clonotypes and decreased TCR clonotype diversity (Suppl. Fig. 2, Suppl. Table 2) suggested that c3 represented an immature $\gamma\delta$ T1 population. The CD24^{low/pos}CD73^{low/pos}44^{low/pos}CD45RB^{low/pos} c4 cluster was enriched in *Ms4a4b* and *Ms4a6b* (encoding CD20 homologues that modulate T cell activation³², ³³), *S1pr1*, *Il6ra*, *Cd8a*, *Cd8b1*, and *Ccr9*, suggesting that the c4 cluster represents the previously described emigrating thymic *Ccr9*⁺ *S1pr1*⁺ $\gamma\delta$ T cells³⁴ (Fig. 1C and 1E). Consistent with the c4 cluster representing naïve emigrating thymic $\gamma\delta$ T cells, the diversity of the c4 TCR clonotypes was the highest among the CD24^{low/int} (c0/c9, c8, c3) clusters (Suppl. Fig. 2).

We identified several immature CD24^{high}CD73^{neg} ADT clusters (c1, c2, c7, c5, c6, and c10) that also expressed SLAMF1 and/or SLAMF6 (Fig. 1B and Suppl. Fig. 1). The c1 and c2 clusters were enriched in $\gamma\delta$ T17-associated genes such as *Blk*, *Maf*, *Sox13*, *Ccr2*, *Tcf12*, and *Rorc*^{26, 35–38}, as well as *Sh2d1a* (encoding SAP) and *Ccr9*¹⁰ (Fig. 1E, Suppl. Table 1). Within these two clusters, the CD44^{low/pos}CD45RB^{low} c2 cluster was especially enriched in genes previously implicated in $\gamma\delta$ T cell development such as *Etv5*, *Tox*, *Sox4*, *Tcf7*, *Zbtb16*^{38–42}, *Igfbp4* and *Igf1r* which were recently implicated in Th17 differentiation⁴³, *Tnfrsf11* (RANKL) which regulates Aire⁺ mTEC development⁴⁴, as well as *Themis* and *Tox2* (Fig. 1E, Suppl. Fig. 1). A high level of *Gzma* expression identified c2 as the previously described *Gzma*⁺ $\gamma\delta$ T cell subset³⁴. Interestingly, although enriched in genes associated with a $\gamma\delta$ T17 signature, the c2 cluster was also enriched in $\gamma\delta$ T2-associated *Gata3* and *Il4* transcripts (Fig. 1D – E, Suppl. Table 1). The CD44^{low/pos}CD45RB^{neg} c1 cluster, in contrast, exhibited significantly higher levels of *Il17a*, *Il17f*, *Rorc*, *Maf*³⁶, as well as *Ifngr1*, consistent with the c1 cluster representing immature $\gamma\delta$ T17 cells (Fig. 1D – E, 1G, and Suppl. Fig. 1).

Analysis of the c1 and c2 TCR transcripts revealed that the c2 cluster was notably enriched in TRGV4-expressing cells and that 87% of V γ 4 T cells utilized just 3 TRDV chains (50% TRDV5, 24% TRDV2-2, and 12% TRAV13-4/DV7) (Fig. 1F, Suppl. Table 2). Clonotype analysis revealed a high frequency of semi-invariant TRGV4/TRDV5 and TRGV4/TRDV2-2 clonotypes previously associated with $\gamma\delta$ T17^{45–47} (Suppl. Fig. 2, Suppl. Table 2). In contrast, the c1 cluster possessed comparatively fewer of these TRGV4 clonotypes, and was enriched in TRGV6, TRGV5, and TRGV1 paired with TRDV4 (Suppl. Fig. 2, Suppl. Table 2). Indeed, we noted that the V γ 1 repertoire, which is diverse in adult tissues, was dominated by TRGV1/TRDV4 clonotypes (35% of all TRGV1) in E17 thymus, and just two closely related semi-invariant TRGV1/TRDV4 clonotypes comprised 17.4% of all TRGV1. The location of these clonotypes in the immature $\gamma\delta$ T17-related clusters (Suppl. Fig. 2) suggested that they represent the dominant IL-17-producing V γ 1 clonotypes at E17, which we verified using flow cytometry (Suppl. Fig. 3).

Since the IL-4-producing V γ 1/V δ 6.3 $\gamma\delta$ NKT cell subset has been shown to develop in fetal thymus¹, we asked whether the *Il4*-enriched $\gamma\delta$ T cells in cluster 2 (Fig. 1D) were immature V γ 1 $\gamma\delta$ NKT cells with previously identified canonical V δ 6.3 sequences²¹. Interestingly, while we did identify a small number of V γ 1 T cells with these canonical sequences (encoded by TRAV15-1/DV6-1 or TRAV15N-1) in our dataset, these were not present in the *Il4*-enriched c2 cluster.

The CD44^{pos}CD45RB^{pos} c5 cluster was enriched in *Il2ra*, *Scin*, *Bex6*, *Cd5*, *Slamf6*, *Cd28*, *Hivep3*, and cell-cycle/metabolism genes, consistent with $\gamma\delta$ T cells that have recently developed from DN2/DN3^{34, 48, 49}. Within the c5 and c7 clusters, we noted that while there was expression of $\gamma\delta$ T17-associated genes such as *Blk*, *Maf*, *Sox13*, and *Etv5*, there was relatively little *Rorc* expression. *Rorc* expression became apparent in the c2 cluster (*Rorc*^{low}) and was significantly increased in the c1 cluster (Suppl. Fig. 1). Flow cytometric analysis confirmed BLK and PLZF expression in CD25^{high} E17 V γ 4 T cells, while ROR γ t expression was primarily observed in the CD25^{neg} population, suggesting that BLK is expressed soon after $\gamma\delta$ T cell selection, followed by ROR γ t (Suppl. Fig. 3). These data were consistent with MAF-directed expression of *Rorc*^{34, 36} and suggested a c5 – c7 – c2 – c10 / c1 – c8 $\gamma\delta$ T17 developmental trajectory, which was supported by trajectory inference analysis (Fig. 1H).

Finally, analysis of the CD24^{high}CD73^{neg}CD44^{neg}CD45RB^{int}SLAMF1^{pos}SLAMF6^{pos} c6 population revealed that it was highly enriched in *Cd8a*, *Cd8b1*, *Cd4*, *Rag1*, *Rag2*, *Ptcra*, *Arpp21*, and *Dgkeos*, consistent with cells developing along the $\alpha\beta$ T cell lineage (Figs. 1E, 1G, Suppl. Fig. 1, and Suppl. Table 1)⁵⁰. Analysis of the V γ 1 and V γ 4 ADTs confirmed that c6 cells were not simply contaminating $\alpha\beta$ T cells as both V γ 1- and V γ 4-expressing cells were present in the c6 cluster (Suppl. Fig. 3) and analysis of the c6 TCR clonotypes revealed the presence of TRGV1, GV4, GV5, GV6, and GV7 clonotypes, none of which appeared to be specific to the c6 cluster (Suppl. Table 2).

Since trajectory inference analysis suggested a strong relationship between the *Il2ra*⁺ c5 and c6 clusters (**Fig. 1H**), we conducted differential gene expression analysis between the c5 and c6 from B6 mice. Interestingly, we found that the c6 cluster was highly enriched in *Rorc*, suggestive of a $\gamma\delta$ T17 phenotype, but was mostly lacking in the expression of *Maf*, *Blk*, *Ccr2*, and other canonical $\gamma\delta$ T17-associated transcripts (**Fig. 1E**, **1G**, and **Suppl. Fig. 1**). Indeed, flow cytometric analysis of E17 TCR β ^{neg}TCR δ ^{pos} $\gamma\delta$ T cells confirmed that CD4CD8 DP $\gamma\delta$ T cells exhibited an immature CD44^{low}CD25^{low}MAF^{neg}BLK^{neg}RORYt^{pos} phenotype (**Suppl. Fig. 3**), and that RORYt expression in the absence of BLK or MAF is a characteristic of Lin^{neg} DN thymocytes as they progress from CD44^{low}25^{high} DN3 to CD44^{low}25^{low} DN4 (**Suppl. Fig. 3**). Together, these data suggested that the c6 cluster $\gamma\delta$ T cells represented previously described CD4CD8 DP $\gamma\delta$ T cells that were developing along an alternative $\alpha\beta$ T cell pathway^{51–53}.

SAP regulates $\gamma\delta$ T cell clonotype diversion into both the $\gamma\delta$ T17 and the $\alpha\beta$ T cell developmental pathways

To identify SAP-dependent developmental checkpoints, we next compared the hashtag-separated B6 and B6.SAP^{-/-} E17 $\gamma\delta$ T cell datasets. A comparison of the cluster frequencies between B6 and B6.SAP^{-/-} $\gamma\delta$ T cells revealed significantly decreased frequencies of the CD24^{high}73^{neg}44^{int}45RB^{int}SLAMF1^{pos}SLAMF6^{pos} c2 and c1 clusters in B6.SAP^{-/-} mice, and a striking increase in the frequency of the CD24^{high}73^{neg}44^{neg}45RB^{neg}SLAMF1^{high}SLAMF6^{high} c6 cluster (**Fig. 2A**). A comparison of differentially expressed genes among the immature (c1, c2, c7, c5, c6, and c10) clusters using pseudobulk analysis revealed 234 differentially expressed genes with a p_{adj} threshold < 0.005 and log₂ fold change greater than 0.5, with 148 genes exhibiting decreased expression in B6.SAP^{-/-} $\gamma\delta$ T cells, and 86 genes exhibiting increased expression (**Fig. 2B**, **Suppl. Table 3**). Consistent with the decreased frequencies of the c1/c2 clusters in B6.SAP^{-/-} mice, a significant fraction of the differentially expressed genes that were down-regulated in B6.SAP^{-/-} mice reflected $\gamma\delta$ T17-associated genes (e.g., *Maf*, *Blk*, *Ccr2*, *Sox13*, *Etv5*, etc.) that were present in the c2 and c1 clusters, as well as a number of c1/c2-enriched genes (e.g., *Cpa3*, *Tox2*, *Pdcd1*, *Tmem121*, *Bex6*, and *Scin*) whose importance in $\gamma\delta$ T cell development is unclear (**Fig. 2B–C**, **Suppl. Table 2**). This was not solely due to the decreased frequency of the c2 and c1 clusters in B6.SAP^{-/-} mice, as a closer analysis revealed a decreased number of transcripts for many signature genes (e.g., *Blk*, *Etv5*, *Ccr2*) prevalent in the c2 cluster and to a lesser extent the c7 cluster (**Suppl. Fig. 4**). In contrast, a large number of genes whose expression was significantly increased in B6.SAP^{-/-} mice were those whose expression was mostly confined to the c6 cluster (e.g., *Cd4*, *Cd8b1*, *Ptcra*, *Rag2*) or whose expression was highest in c6 (e.g., *Tcf7*, *Thy1*) (**Fig. 2B–C**, **Suppl. Fig. 4**). Together, these data identified the *Gzma*⁺*Etv5*⁺*Sox13*⁺*Blk*⁺*Tox2*⁺c2 and *Cd4*⁺*Cd8b1*⁺*Ptcra*⁺*Rag2*⁺c6 clusters as SAP-dependent developmental checkpoints.

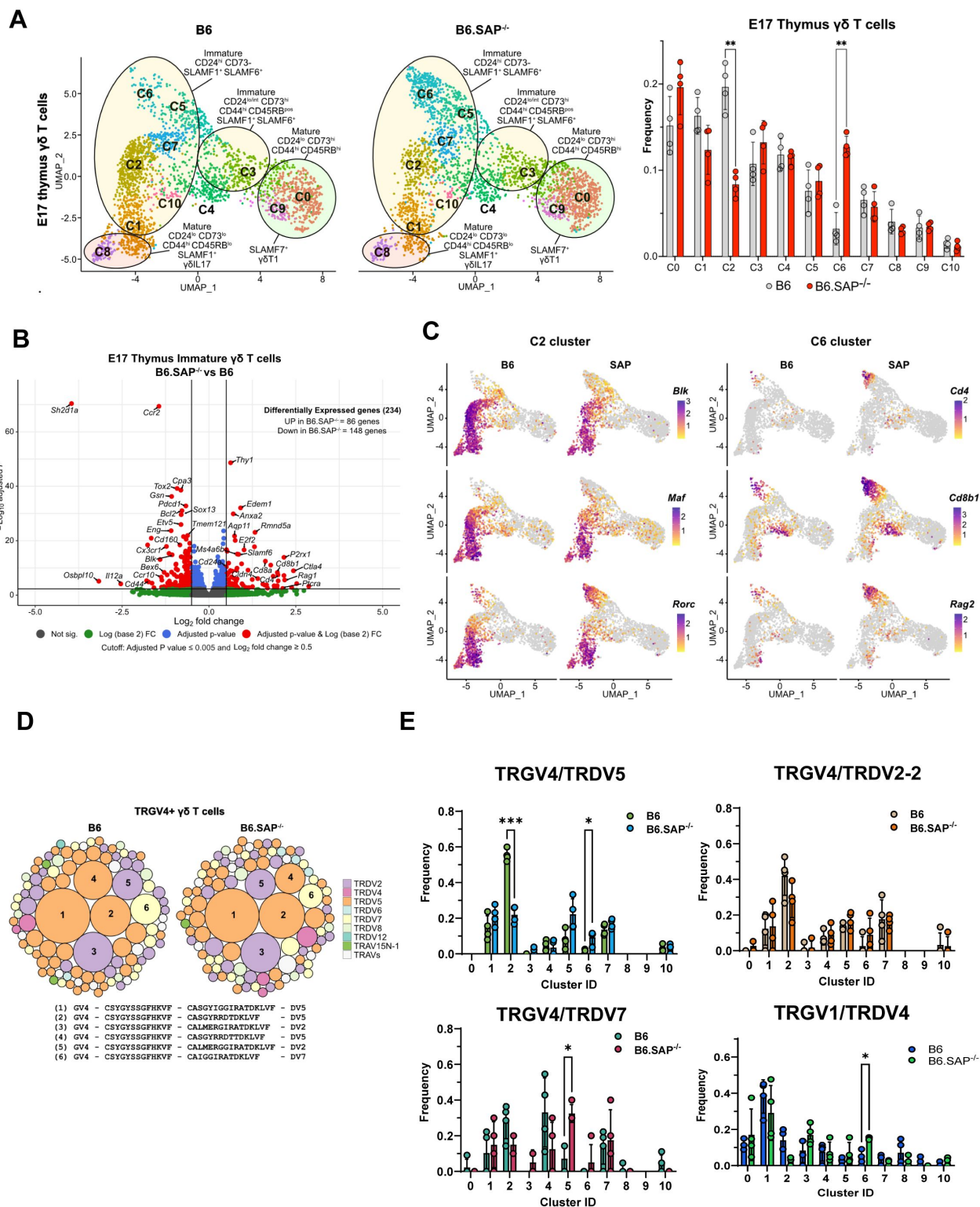


Figure. 2.

Identification of SAP-dependent developmental checkpoints during E17 $\gamma\delta$ T cell developmental programming.

(A) UMAP representation of B6 (left) and B6.SAP^{-/-} (right) $\gamma\delta$ T cells from E17 thymi (n = 4 mice per group) is shown at left. Clusters were annotated based on protein and gene expression data. The frequencies of B6 and B6.SAP^{-/-} E17 $\gamma\delta$ T cell clusters is shown at right. Bars represent the mean cluster frequency, error bars represent standard deviation, **p ≤ 0.01, two-way ANOVA, Sidak multiple-comparisons test; n=4 mice/genotype. (B) Volcano plot of differentially expressed genes among immature CD24^{high}CD73^{neg} (clusters: C1, C2, C5, C6, C7, and C10) B6 and B6.SAP^{-/-} $\gamma\delta$ T cells using pseudobulk scRNAseq analysis. Genes exhibiting a log₂FC ≥ 0.5 or ≤ -0.5, and a p_{adj} ≤ 0.005 in are shown in red. Only some selected genes are labelled for the sake of clarity. (C) Feature plots illustrating gene expression profiles of selected C2 and C6 cluster-specific genes among individual B6 and B6.SAP^{-/-} E17 thymic $\gamma\delta$ T cells. Each point represents a cell, color-coded by gene expression level (high: purple, low: yellow). (D) TCR clonotype bubble plot displaying the top 100 B6 and B6.SAP^{-/-} TRGV4 clonotypes in E17 thymus. Bubbles represent unique TRGV4/TRDV clonotypes; size indicates frequency of clonotype among all V γ 4 T cells (n = 388 B6, 288 B6.SAP^{-/-} cells), and colors denote specific TRDV chains utilized. Selected clonotypes are numbered and their respective CDR3 γ and CDR3 δ sequences displayed below. Data are (E) Altered TRGV/TRDV clonotype distribution between B6 and B6.SAP^{-/-} E17 thymic $\gamma\delta$ T cells. Frequencies of selected clonotypes among different clusters of B6 and B6.SAP^{-/-} E17 $\gamma\delta$ T cells is shown. Bars represent the mean clonotype frequency among the indicated TRGV/TRDV pairings shown, error bars represent standard deviation, *p ≤ 0.05, ***p ≤ 0.001, two-way ANOVA, Sidak multiple-comparisons test; n = 4 mice/genotype-age groups.

Next, we asked whether these SAP-dependent alterations in $\gamma\delta$ T cell development were associated with any changes in the TCR repertoire. A comparison of all $\gamma\delta$ TCR clonotypes between E17 B6 and B6.SAP^{-/-} $\gamma\delta$ T cells revealed no significant changes in overall clonotype frequency (Fig. 2D). However, a closer examination of TCR repertoire revealed significant changes in the distribution of specific TCR clonotypes. For example, within the TRGV4 repertoire, which represents the dominant TCR among immature clusters at E17, we observed a significant decrease in TRGV4/TRDV5 clonotypes in the c2 cluster and a concomitant increase of these clonotypes in the c6 cluster (Fig. 2E). Among the TRGV4 clonotypes, this appeared to be specific to the TRDV5 clonotypes, as we observed no significant differences c2 and c6 frequencies among TRGV4/TRDV2-2 and TRDV7 clonotypes (Fig. 2E). We did note, however, a significant increase in the TRGV4/TRDV7 clonotypes in the *Il2ra*⁺ c5 cluster (Fig. 2E). Within the TRGV1 repertoire, we noted a significant increase in TRGV1/TRDV4 clonotypes within the c6 cluster (Fig. 2E). Together, these data suggested that SAP operated at an early stage of $\gamma\delta$ T cell development during or soon after the down-regulation of *Il2ra*, and that it positively regulated entry into the $\gamma\delta$ T17 pathway at the c5/c7 to c2 transition, while at the same time inhibiting entry into the CD4⁺CD8⁺ $\alpha\beta$ T cell-like c6 cluster. The net result of these changes appeared to be that specific TCR clonotypes such as TRGV4/TRDV5 were diverted from the $\gamma\delta$ T17 pathway into the $\alpha\beta$ T cell-like pathway.

SAP deficiency is associated with increased numbers of immature CD4⁺CD8⁺ROR γ t⁺ thymic $\gamma\delta$ T cells

Next, we independently evaluated these observations using flow cytometry. This analysis revealed that while SAP deficiency did not affect the overall number of E17 $\gamma\delta$ T cells (Suppl. Fig. 5), it did result in a significant decrease of immature CD24^{pos}BLK^{pos}MAF^{pos}ROR γ t^{pos} $\gamma\delta$ T cells in E17 B6.SAP^{-/-} thymus (Fig. 3A-B). We found that this decrease was most pronounced in the CD44^{pos} population after downregulation of CD25 (Fig 3A, Suppl. Fig. 5) and that this population exhibited a high level of PLZF expression, suggesting it corresponded to the CD44⁺Blk⁺Maq⁺Zbtb16⁺Rorc⁺ c1 cluster (Suppl. Fig. 5). In addition, UMAP analysis of E17 V γ 4 T

cells flow cytometric data revealed the presence of a SAP-dependent CD24^{pos}BLK^{pos}PLZF^{pos}RORγt^{neg} population in B6.SAP^{-/-} thymus (Suppl. Fig. 5 [↗](#)) consistent with the phenotype of the c2 cluster which expressed lower levels of *Rorc*. In contrast, we found no evidence of a SAP-dependent decrease in the CD25^{pos} population frequency, consistent with the notion that SAP exerts its effect during or after this stage of development (Fig. 3A [↗](#)). We also noted that in addition to its effect on the numbers of immature thymic γδT17, SAP deficiency resulted in the reduced expression levels of some of these markers, most notably BLK (Fig. 3C [↗](#)), a finding that was consistent with the decreased number of *Blk* transcripts observed in Fig. 2 [↗](#). Interestingly, as PLZF induction is often associated with SLAM/SAP signaling, we noted that SAP did not affect overall PLZF expression levels in E17 thymic γδ T cells (Suppl. Fig. 5 [↗](#)).

In contrast to the decreased numbers of CD24^{pos}BLK^{pos}MAF^{pos}PLZF^{pos} RORγt^{pos} γδ T cells, we observed a significant increase in the frequency and number of CD24^{pos}BLK^{neg}MAF^{neg}PLZF^{neg}RORγt^{pos} γδ T cells (Figs. 3A-B [↗](#)), whose phenotype was consistent with the αβ T cell-like c6 cluster in Fig. 2 [↗](#). Consistent with our finding that the c6 cluster was enriched in CD4 and CD8, we found that the frequency and number of CD4 and CD8-expressing CD24^{pos} TCRδ^{pos}TCRβ^{neg} BLK^{neg}MAF^{neg}RORγt^{pos} thymic γδ T cells was significantly increased in B6.SAP^{-/-} mice (Fig. 3D [↗](#)). Together, these analyses confirmed our single-cell CITEseq analysis, and suggested that SAP regulated both the throughput of immature γδ T cells in the γδT17 cell pathway as well as the αβ T cell-like pathway. More specifically, the data suggested that SAP promoted progression to the CD24^{pos} BLK^{pos}MAF^{pos}PLZF^{pos}RORγt^{low} c2 cluster that is enriched in a number of genes (e.g., *Sox13*, *Etv5*, *Blk*, *Maf*) known to regulate γδT17 development, but that it inhibited entry into the CD24^{pos}44^{neg}BLK^{neg}MAF^{neg}RORγt^{pos} αβ-T cell-like pathway (c6 cluster).

Identification of SAP-dependent developmental checkpoints during neonatal and adult thymus γδ T cell developmental programming

The neonatal period is a time of rapid expansion of numerous γδ T cell subsets in the thymus including Vγ4 γδT17 and γδNKT. Therefore, we conducted a comparison between B6 and B6.SAP^{-/-} neonatal as well as adult thymic γδ T cells to both identify novel SAP-dependent developmental checkpoints and to track the outcome of the SAP-dependent alterations in γδ T cell development observed in E17 thymus. Neonate datasets revealed a similar branched clustering pattern as in E17 with some notable differences. Similar to E17, we observed immature CD24^{pos}73^{low}SLAMF1^{pos}SLAMF6^{pos} clusters (c0, c1, c2, c3, c9, c12, c14), CD24^{pos}73^{low/pos}SLAMF1^{pos}SLAMF6^{pos} (c4, c5, c10, c15) *S1pr1*-enriched “naïve” clusters, mature CD24^{neg}73^{low}SLAMF1^{pos}SLAMF6^{neg} (c6, c7) gdT17 clusters, and a mature CD24^{neg}73^{pos}SLAMF1^{neg}SLAMF6^{low/pos}*Slamf7*^{high} (c8) gdT1 cluster (Fig. 4A-B [↗](#), Suppl. Fig. 6 [↗](#)). In general, we noted good correlation with cluster-specific gene expression between the E17 and D9 (Day 9) datasets. A notable exception was a striking decrease in *Zbtb16* expression among all immature neonatal γδ T cells, which is consistent with previous reports^{10 [↗](#), 34 [↗](#)} (Suppl. Fig. 6 [↗](#), Suppl. Table 3).

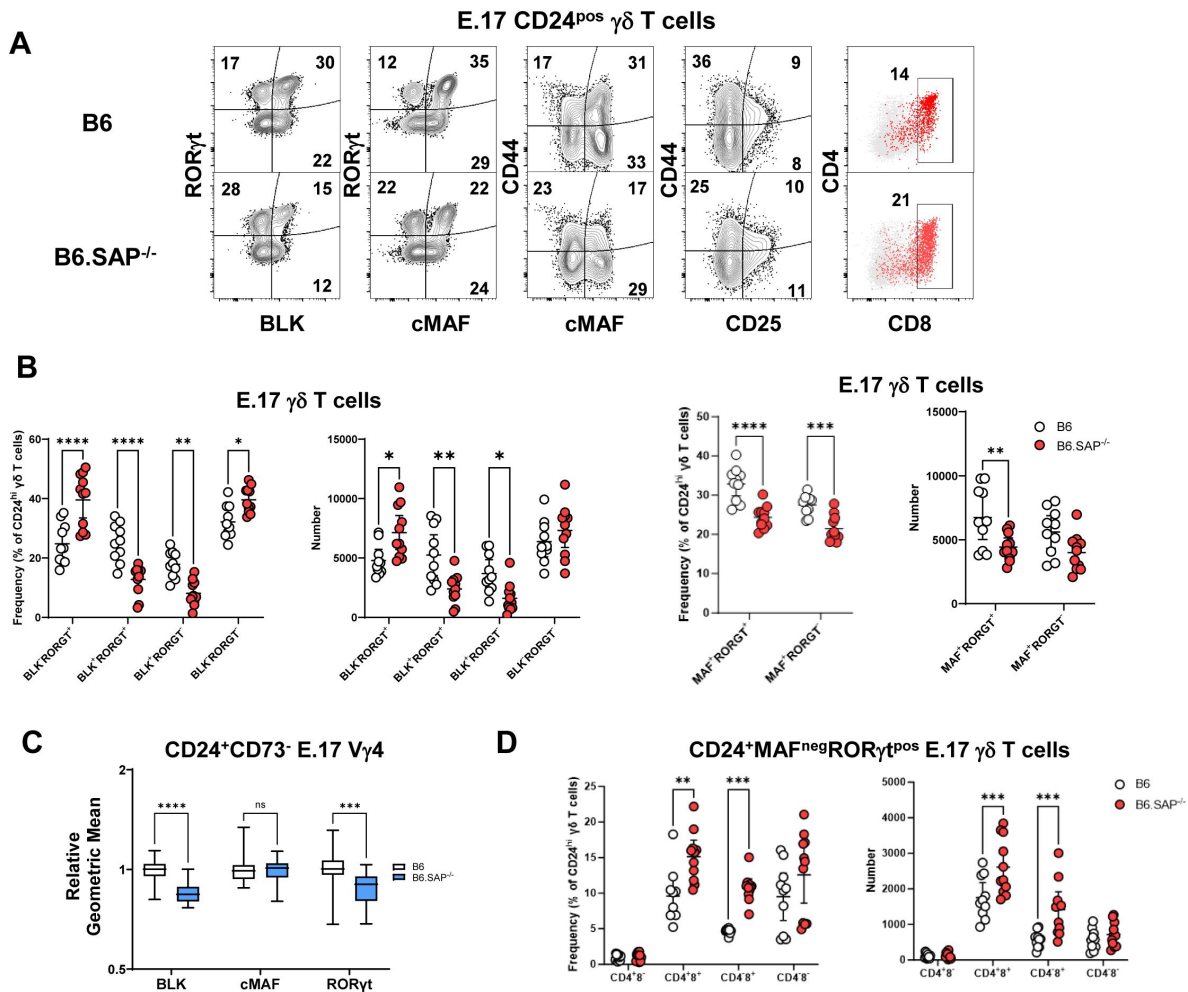


Figure 3.

Increased frequency and numbers of immature CD4⁺CD8⁺c-MAF⁺RORγt⁺ thymic γδ T cells in B6.SAP^{-/-} E17 γδ T cells.

(A) Representative contour plots depicting CD24^{pos} E17 γδ T cells in B6 (top) and B6.SAP^{-/-} (bottom) mice. Concatenated data from 2 B6 and 3 B6.SAP^{-/-} embryos are shown, and are representative of 2 independent experiments. Numbers in the plots represent the frequency as a percentage of CD24^{pos} γδ T cells. (B) Cumulative frequency and number of immature CD24^{pos} BLK⁻ (left), c-MAF⁻ (right), RORγt-expressing E17 thymic γδ T cells. The mean and standard deviation are indicated. Data are the cumulative data from 2 independent experiments, n = 10 B6, 11 B6.SAP^{-/-} pooled embryonic thymi, *p ≤ 0.05, **p ≤ 0.01, ***p ≤ 0.001, ****p ≤ 0.0001, two-way ANOVA, Sidak's multiple comparisons test. (C) Relative geometric mean of BLK, c-MAF, and RORγt expression in B6 and B6.SAP^{-/-} E17 CD24⁺CD73⁻ Vγ4 γδ T cells, *p ≤ 0.001, ****p ≤ 0.0001, two-way ANOVA, Sidak's multiple comparisons test. Data are representative of 2 independent experiments, 7-9 mice per group. (D) Cumulative frequency (left) and number (right) of immature CD4⁻ and CD8⁻ MAF^{neg}RORγt^{pos} E17 thymic γδ T cells in B6 and B6.SAP^{-/-} thymus. The mean and standard deviation are indicated. Data are the cumulative data from 2 independent experiments, n = 10 B6, 11 B6.SAP^{-/-} pooled embryonic thymi, **p ≤ 0.01, ***p ≤ 0.001, ****p ≤ 0.0001, two-way ANOVA, Sidak's multiple comparisons test.

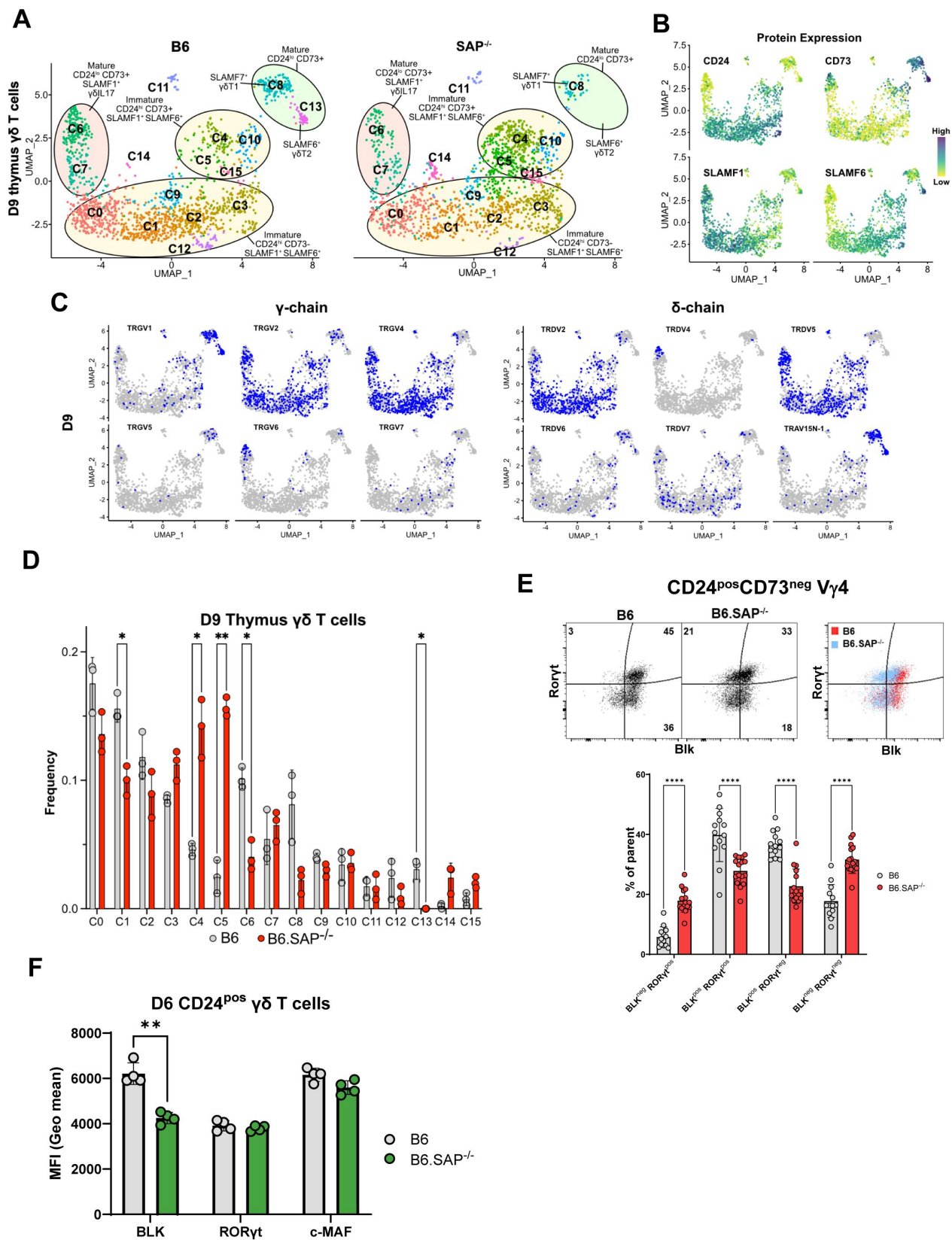


Figure 4.

Identification of SAP-dependent developmental checkpoints during neonatal $\gamma\delta$ T cell developmental programming.

(A) UMAP representation of B6 (left) and B6.SAP^{-/-} (right) $\gamma\delta$ T cells from D9 neonatal thymi (n = 3 mice per strain). Cluster annotation is based on comprehensive protein and gene expression data. (B) Feature plots displaying cell surface protein expression patterns of CD24, CD73, SLAMF1, and SLAMF6 on D9 B6 thymus $\gamma\delta$ T cells. Data are color-coded based on protein expression level (high: dark blue, low: yellow). (C) UMAP representation of D9 B6 thymic $\gamma\delta$ T cells exhibiting selected TRGV4 (left) and TRDV (right) chain V-segment usage (in blue) in individual cells. (D) Frequencies of B6 and B6.SAP^{-/-} D9 thymic $\gamma\delta$ T cell clusters. Bars represent the mean cluster frequency, error bars represent standard deviation, *p ≤ 0.05, **p ≤ 0.01, two-way ANOVA, Sidak multiple-comparisons test, 3 mice/strain. (E) Representative contour plots depicting BLK and RORyt expression in B6 and B6.SAP^{-/-} CD24^{pos}CD73^{neg} D9 V γ 4 T cells are shown *above*. Cumulative frequencies of BLK and RORyt expression are shown *below*. The mean and standard deviation are indicated. Data are the cumulative data from 3 independent experiments, n = 13 B6, 16 B6.SAP^{-/-} mice, ****p ≤ 0.0001, two-way ANOVA, Sidak's multiple comparisons test. (F) Decreased BLK expression in immature B6.SAP^{-/-} $\gamma\delta$ T cells. Data represent the mean expression of the indicated proteins in D6 neonate BLK^{pos}, RORyt^{pos}, or c-Maf^{pos} thymic $\gamma\delta$ T cells, **p ≤ 0.01.

In contrast to the E17 dataset, the neonatal thymic $\gamma\delta$ T cell dataset contained a CD24^{neg}73^{pos}SLAMF1^{neg}SLAMF6^{high}Slamf7^{neg} (c13) $\gamma\delta$ T2 cluster enriched in *Il4*, *Il13*, *Zbtb16*, *Cd40lig*, and *Nos1* (Suppl. Fig. 6, Suppl. Table 1), suggesting it represented IL-4-producing $\gamma\delta$ NKT cells. Indeed, examination of the $\gamma\delta$ T2 (c13) and $\gamma\delta$ T1 (c8) cluster TCR repertoires revealed a significant enrichment of canonical TRGV1 and TRAV15N-1 or TRAV15-1/DV6-1 sequences characteristic of $\gamma\delta$ NKT cells (Fig. 4C, Fig. 5).²¹ Analysis of the differentially expressed genes in the $\gamma\delta$ T1 (c8) and $\gamma\delta$ T2 (c13) clusters revealed that *Zbtb16*, *Icos*, and *Slamf6* were markers of $\gamma\delta$ T2 cells while *Tbx21*, *Klrb1c* (NK1.1), and *Slamf7* were markers of $\gamma\delta$ T1 cells in neonatal thymus (Suppl. Fig. 6). Flow cytometric analysis confirmed that PLZF^{hi} $\gamma\delta$ T2 V γ 1V δ 6.3 cells co-express SLAMF6 and ICOS while Tbet^{hi} $\gamma\delta$ T1 V γ 1V δ 6.3 cells are NK1.1⁺ and are either SLAMF6⁻SLAMF7⁺ or SLAMF6⁺SLAMF7⁺ (Suppl. Fig. 7).

A comparison of B6 and B6.SAP^{-/-} neonatal gd T cells revealed multiple SAP-dependent changes in clustering. Among the immature $\gamma\delta$ T cells, we observed a decreased frequency of the immature CD24^{high}73^{neg}SLAMF1^{pos}SLAMF6^{pos} c1 cluster (Fig. 4D) that was enriched in *Blk*, *Etv5*, *Sox13*, *Rorc*, and *Maf* (Suppl. Fig. 6), suggesting it was analogous to the SAP-dependent c2 cluster in E17 $\gamma\delta$ T cells. Similar to our observations in E17 $\gamma\delta$ T cells, flow cytometric analysis confirmed both a decrease in BLK^{pos}RORyt^{pos} and BLK^{pos}RORyt^{neg} $\gamma\delta$ T cell populations, as well as a decreased level of BLK expression (Fig. 4E-F). In addition, we observed an increased frequency of immature CD24^{high}73^{neg}SLAMF1^{pos}SLAMF6^{pos} cluster with characteristics of $\alpha\beta$ T cells (c14) in B6.SAP^{-/-} D9 thymic $\gamma\delta$ T cells (Fig. 4D, Suppl. Fig. 6). While this increase did not reach the level of significance, presumably due to the very low number of c14 cells in the dataset, flow cytometric analysis revealed a significant increase in immature CD24^{pos}73^{neg}BLK^{neg}RORgt^{pos} gd T cells, a phenotype consistent with gd T cells that have been diverted to the ab developmental pathway (Fig. 4E). Last, we observed an increased frequency of immature CD24^{pos}73^{low/pos}SLAMF1^{pos}SLAMF6^{pos} *S1pr1*-enriched c4 and c5 clusters, but not the closely related *Cd200*-enriched c10 cluster (Fig. 4D). While we did confirm an increased number CD200^{neg}S1pr1^{pos} $\gamma\delta$ T cells in neonatal thymus by flow cytometry, this increase was relatively small and its significance is unclear (Suppl. Fig. 7). Independent confirmation of these SAP-dependent changes in neonatal $\gamma\delta$ T cell gene expression was obtained by conducting bulk RNAseq analysis of FACS-sorted CD24^{high} neonatal thymic $\gamma\delta$ T cells from B6 and B6.SAP^{-/-} mice (Suppl. Table 4).

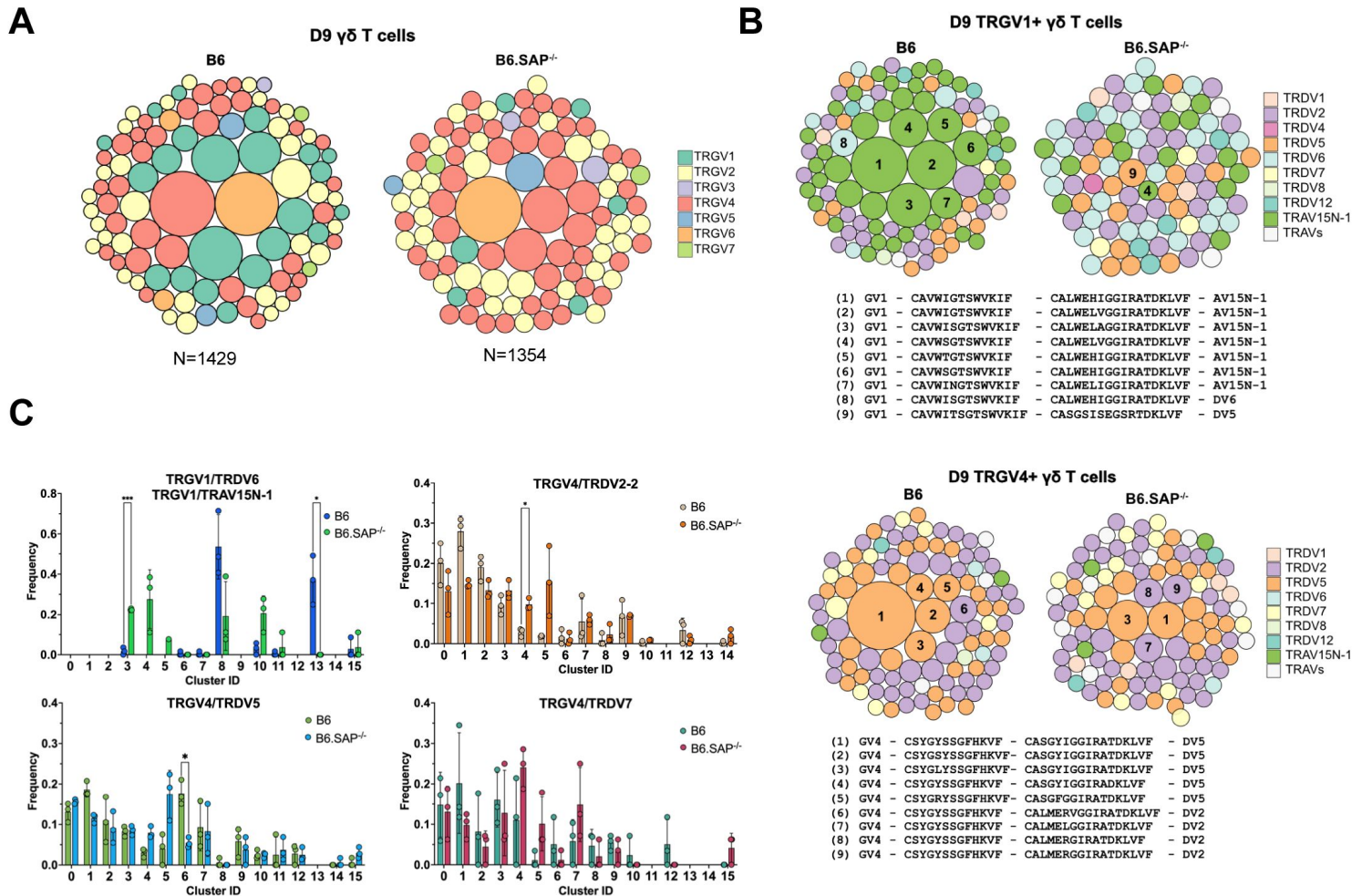


Figure 5.

SAP shapes the neonatal $\gamma\delta$ TCR repertoire.

(A) TCR clonotype bubble plots depicting the top 100 B6 and B6.SAP^{-/-} TCR clonotypes among D9 thymic $\gamma\delta$ T cells. Each bubble represents a unique clonotype, is sized according to its frequency as a percentage of all $\gamma\delta$ T cells ($n = 1429$ B6, 1354 B6.SAP^{-/-} cells), and is colored based on specific TRGV chains utilized. **(B)** TCR clonotype bubble plots depicting the top 100 B6 and B6.SAP^{-/-} TRGV1 (above) and TRGV4 (below) clonotypes among D9 $\gamma\delta$ T cells. Bubble size indicates clonotype frequency as a percentage of TRGV1⁺ ($n = 198$ B6, 98 B6.SAP^{-/-}) or TRGV4⁺ ($n = 646$ B6, 669 B6.SAP^{-/-}) cells, and colors correspond to specific TRDV chains. Selected clonotypes are numbered and their corresponding CDR3 γ and CDR3 δ sequences are displayed below. **(C)** Distribution of selected TRGV/TRDV clonotypes among B6 and B6.SAP^{-/-} D9 thymic $\gamma\delta$ T cell clusters. Data representative of 3 mice/strain, * $p \leq 0.05$, *** $p \leq 0.001$, two-way ANOVA, Sidak multiple-comparisons test.

Among the mature neonatal $\gamma\delta$ T cells, we observed a decreased frequency of the $CD24^{low}73^{neg}SLAMF1^{pos}F6^{low}$ $\gamma\delta T17$ c6 cluster which was consistent with our previous observation of decreased IL-17 production in $B6.SAP^{-/-}$ neonatal $\gamma\delta$ T cells¹⁸. Interestingly, we observed a striking loss of the $CD24^{low}73^{high}SLAMF1^{low}F6^{high}F7^{neg}$ $\gamma\delta T2$ c13 cluster in $B6.SAP^{-/-}$ mice, as well as a decreased frequency of the mature $CD24^{low}73^{high}SLAMF1^{low}F6^{low}F7^{high}$ $\gamma\delta T1$ c8 cluster (**Fig. 4A - D**). These findings were consistent with a SAP-dependent loss of both the IL-4- and IFN- γ -producing $\gamma\delta$ NKT as well as non- $\gamma\delta$ NKT $\gamma\delta T1$ cells (discussed further below). Unfortunately, while we were able to define mature SAP-dependent $\gamma\delta$ NKT subsets, we found no evidence of immature $\gamma\delta$ NKT cells in our dataset, suggesting that they develop during a brief developmental window between E17 and D9 after birth.

A comparison of B6 adult thymic $\gamma\delta$ T cells between B6 and $B6.SAP^{-/-}$ mice recapitulated many of our observations of the neonatal thymus, although to a somewhat lesser degree. For example, we observed a small but significant decrease in the frequency of the $CD24^{high}CD73^{low}$ adult thymic c1 and c2 clusters in $B6.SAP^{-/-}$ mice that was coupled with an increased frequency of the $CD24^{pos}CD73^{+/-}$ c0 cluster (**Suppl. Fig. 8**). Like the SAP-dependent E17 c2 and D9 c1 clusters, the adult thymus c1 and c2 clusters were enriched in $\gamma\delta T17$ -associated *Blk*, *Etv5*, *Sox13*, *Maf*, and *Rorc* (mostly in c2) indicating that SAP regulates this $\gamma\delta$ T cell thymic developmental checkpoint at all ages. Likewise, the adult thymus c0 cluster that was increased in $B6.SAP^{-/-}$ mice was enriched in *Ms4a4b*, *Ms4a6b*, *Dgka*, *Klf2*, and *S1pr1* similar to the neonatal c4 and c5 clusters whose frequencies were increased in $B6.SAP^{-/-}$ $\gamma\delta$ T cells. Just as we observed in neonatal thymic $\gamma\delta$ T cells, we noted a near complete loss of mature $\gamma\delta$ NKT cells (*Zbtb16*, *Icos*, and *Il4*-enriched $\gamma\delta T2$ c8 cells and *Tbx21*, *Slamf7*, and *Il2rb*-enriched $\gamma\delta T1$ c11 cells) as well as a decreased frequency of mature non- $\gamma\delta$ NKT $\gamma\delta T1$ c10 cluster cells in $B6.SAP^{-/-}$ adult thymus (**Suppl. Fig. 8**). Unlike the E17 and neonatal thymic datasets, however, the frequency of $\gamma\delta$ T cells with an $\alpha\beta$ T cell-like signature in the adult thymic $\gamma\delta$ T cell dataset was minimal. Collectively, these data supported the presence of multiple SAP-dependent developmental checkpoints during neonatal and adult $\gamma\delta$ T cell development—a reduction in the $CD24^{pos}CD73^{neg}SLAMF1^{+}SLAMF6^{+}$ cluster enriched in $\gamma\delta T17$ development-associated transcripts, an increased frequency of immature *S1pr1*-enriched $\gamma\delta$ T cells, and decreased frequencies of both $\gamma\delta$ NKT cells ($\gamma\delta T1$ and $\gamma\delta T2$) as well as non- $\gamma\delta$ NKT $\gamma\delta T1$ cells.

SAP shapes the thymic $\gamma\delta$ TCR repertoire

Given our observation of altered $\gamma\delta$ TCR clonotype distribution among E17 $\gamma\delta$ T cells and the substantial changes in clustering we observed in SAP-deficient mice, we were next interested in determining the effect of SAP deficiency on the neonatal and adult $\gamma\delta$ TCR repertoires. A comparison of frequencies of the top 100 TCR clonotypes between B6 and $B6.SAP^{-/-}$ mice revealed significant changes in the overall frequency of several TRGV1 and TRGV4 clonotypes (**Fig. 5A**). Consistent with a SAP-dependent loss of $\gamma\delta$ NKT, we observed a profound loss of TRGV1 clonotypes utilizing the canonical TRAV15N-1 and TRAV15-1/TRDV6-1 $\gamma\delta$ NKT cell TCRs (**Fig. 5B**). Loss of these clonotypes was directly tied to the significant decrease of the $\gamma\delta T1$ c8 and $\gamma\delta T2$ c13 clusters, in which nearly all these clonotypes resided.

Within the B6 $\gamma\delta T2$ c13 cluster the vast majority (~86%) of TCR sequences were highly restricted canonical $\gamma\delta$ NKT cell clonotypes (**Suppl. Table 2**) and nearly all of these clonotypes were also present in the $\gamma\delta T1$ c8 cluster consistent with the ability of $\gamma\delta$ NKT cells to produce both IFN- γ and IL-4. In contrast, while the B6 $\gamma\delta T1$ c8 cluster contained a large proportion of canonical $\gamma\delta$ NKT cell clonotypes (~46% of c8 TCRs), it also contained TCRs with a diverse pool of clonotypes that utilized TRGV1 (~70% of c8 TCRs), TRGV4, TRGV5, TRGV6, and TRGV7 and were present only in the c8 $\gamma\delta T1$ cluster. Therefore, in neonates, the thymic $\gamma\delta T1$ population is a mixture of $\gamma\delta$ NKT and “non- $\gamma\delta$ NKT” $\gamma\delta T1$ clonotypes, and each of these clonotype pools is diminished in $B6.SAP^{-/-}$ mice. As noted above, these findings were largely reproduced in the adult thymus where the $\gamma\delta$ NKT clonotypes associated with the c8 and c11 clusters were virtually absent in $B6.SAP^{-/-}$ mice, and where the diverse pool of $\gamma\delta T1$ c10 clonotypes was significantly diminished (**Suppl. Table 2**).

Interestingly, a closer examination of the neonatal thymic TRGV4 clonotypes revealed a significant decrease in the frequencies of $\gamma\delta$ T17-associated clonotypes with a germline-encoded TRDV5⁴⁶ (Fig. 5B), some of which were identical to those that were redistributed to the $\alpha\beta$ T cell-like c6 cluster in E17 $\gamma\delta$ T cells (Fig. 2E). When these clonotypes were mapped to their D9 clusters, we noted a significant decrease in the frequency of TRGV4/TRDV5 clonotypes in the mature $\gamma\delta$ T17 c6 cluster (Fig. 5C). We note that while we observed a trend toward an increased frequency of TRGV4/TRDV5 and TRGV4/TRDV2 clonotypes in the *S1pr1*⁺ c4/c5 clusters, analysis of these sequences revealed that they largely exhibited diverse CDR3 rather than the restricted CDR3 observed in the $\gamma\delta$ T17 c6 cluster (Suppl. Table 2). Taken together, these data indicate that the TCR clonotypes of immature embryonic day 17 $\gamma\delta$ T cells that were redirected to the $\alpha\beta$ T cell pathway in B6.SAP^{-/-} mice, were underrepresented among the mature B6.SAP^{-/-} neonatal $\gamma\delta$ T cells.

Restricted TCR repertoire in innate-like $\gamma\delta$ T1 as well as $\gamma\delta$ T17 in the periphery

We previously demonstrated that SAP-deficiency results in a significant impairment in $\gamma\delta$ T cell function in the periphery¹⁸. Specifically, we observed that $\gamma\delta$ T cell IL-17 production is impaired in the lungs of young mice, but that this had largely corrected itself as the mouse reached maturity. In contrast, we observed that both V γ 1 and V γ 4 IFN- γ production were significantly impaired in B6.SAP^{-/-} adult lung and spleen. To better understand how SAP-dependent alterations in thymic development ultimately affected the peripheral $\gamma\delta$ T cell compartment, we compared B6 and B6.SAP^{-/-} lungs using single-cell RNAseq with V(D)J profiling.

Our initial annotation of the B6 lung $\gamma\delta$ T cell transcriptome revealed the presence of 11 clusters of the lung $\gamma\delta$ T cells that we annotated based on cluster-specific gene expression and TRGV and TRDV usage (Figure 6A). Based on their expression of signature cytokines and transcription factors, we identified 2 $\gamma\delta$ T17 clusters (c1 and c4), 4 $\gamma\delta$ T1 clusters (c2, c3, c7, and c9), 1 $\gamma\delta$ TNKT-like cluster (c10) with very few cells, as well as a *Cd24*-enriched cluster (c5) (Fig. 6B and 6C, Suppl. Table 5). Consistent with previous observations, we noted that the c1 and c4 $\gamma\delta$ T17 clusters were enriched in *Slamf1* and that the c2/c3/c7/c9 $\gamma\delta$ T1 clusters were enriched in *Slamf7* and *Slamf6* (Fig. 6B). Indeed, flow cytometric analysis revealed that SLAMF7 expression was restricted to the innate-like lung CD44⁺CD45RB⁺ $\gamma\delta$ T1 cell population, and that $\gamma\delta$ T1 cells are SLAMF1⁺F6⁺F7⁺, while the CD44⁺CD45RB⁻ $\gamma\delta$ T17 subsets are SLAMF1⁺F6⁻F7⁻ (Fig. 6D). Further annotation using V(D)J profiling revealed that the c1 cluster corresponded to V γ 4 $\gamma\delta$ T17 (primarily TRGV4/TRDV5 and TRGV4/TRDV2-2) while the c4 cluster corresponded to lung V γ 6 (TRGV6/TRDV4) (Fig. 6E, Suppl. Table 6). The c10 cluster was enriched in *Zbtb16*, *Il4*, and *Icos* (Fig. 6B-C) as well as TRGV1/TRAV15N-1 and TRGV1/TRDV6 sequences (Fig. 6E, Suppl. Table 6) consistent with $\gamma\delta$ NKT cells. We did note, however, that the TRAV15N-1 and TRDV6 sequences in the c10 cluster were, for the most part, non-canonical $\gamma\delta$ NKT cell clonotypes, suggesting a degree of flexibility in the TRAV15N-1 and TRDV6 CDR3 sequences that can be utilized by $\gamma\delta$ NKT cells.

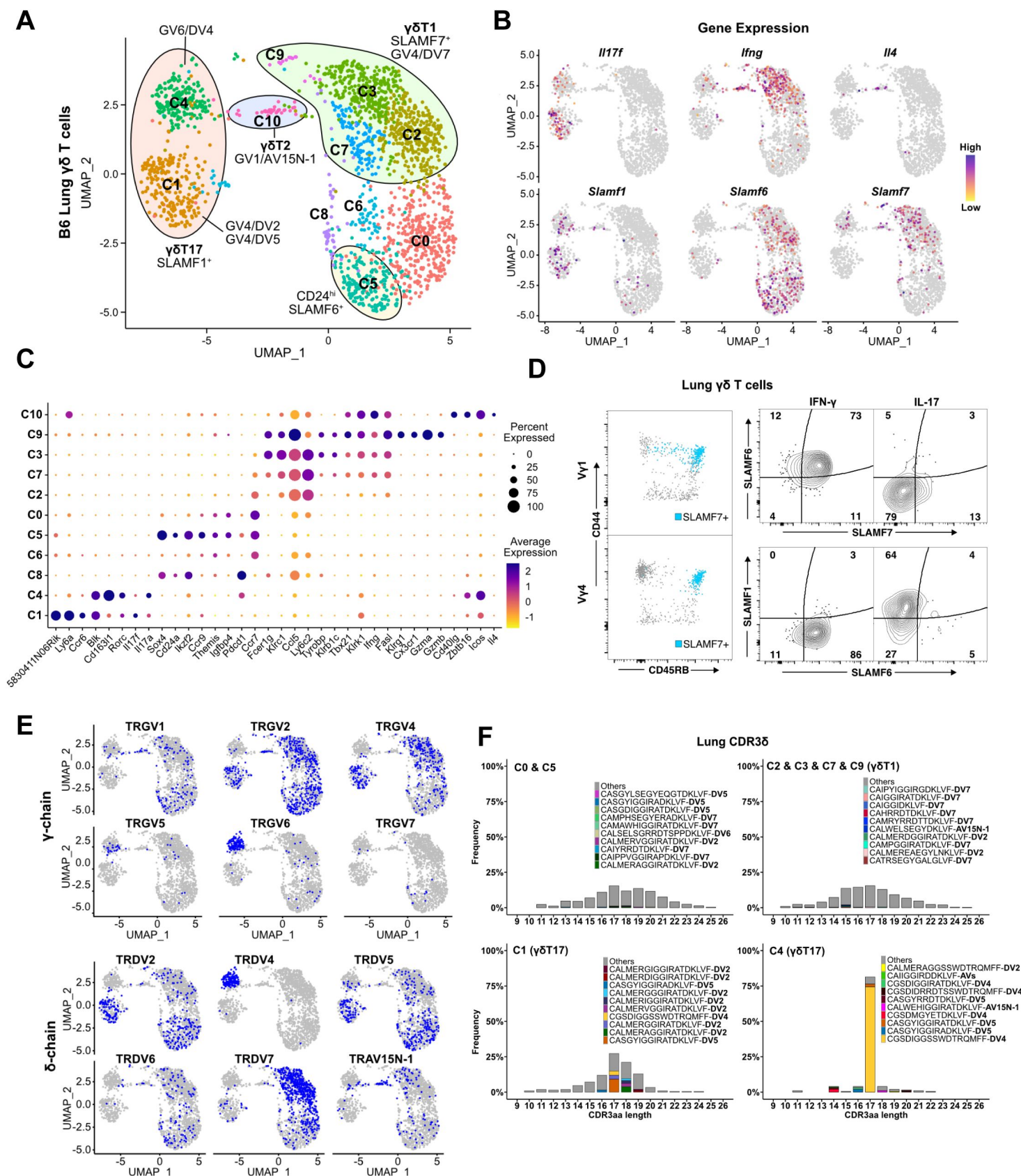


Figure 6.

Restricted TCR repertoire in innate-like $\gamma\delta$ T1 as well as $\gamma\delta$ T17 in the periphery.

(A) UMAP representation of B6 adult lung $\gamma\delta$ T cells. Clusters are annotated based on gene expression and TCR profiling data. (B) Feature plot displaying gene expression profiles of selected genes among individual B6 lung $\gamma\delta$ T cells, color-coded based on gene expression levels. (C) Dot plot indicating scaled expression levels of selected genes in B6 lung $\gamma\delta$ T cells, with dot size representing the fraction of cells within each cluster expressing the marker. (D) SLAMF6 and SLAMF7 co-expression marks CD44⁺CD45RB⁺ IFN- γ -producing cells in the periphery. Representative dot plots (left) of CD44 and CD45RB expression on lung V γ 1 (top) and V γ 4 (bottom) $\gamma\delta$ T cells. SLAMF7-expressing cells are highlighted in blue. Representative contour plots (right) of SLAMF1, SLAMF6, and SLAMF7 expression on IFN- γ ⁺ and IL-17⁺ lung $\gamma\delta$ T cells. (E) UMAP representation of B6 lung $\gamma\delta$ T cells indicating selected TRG and TRD chain V-segment usage (blue). (F) Amino acid length distribution of lung CDR3 δ clonotypes among immature C0/C5, $\gamma\delta$ T1 C2/C3/C7/C9, $\gamma\delta$ T17 C1, and $\gamma\delta$ T17 C4 clusters. The top 10 clonotypes in each group are color-coded, while all other clonotypes are shown in gray.

Interestingly, while the $\gamma\delta$ T1 c2/c3/c7/c9 clusters contained primarily V γ 1 and V γ 4 T cells as expected, we noted that V γ 4 $\gamma\delta$ T1 cells exhibited a striking preference in the usage of TRAV13-4(DV7), and that TRGV4/TRDV7 pairings accounted for 80% of all V γ 4 clonotypes in the $\gamma\delta$ T1 clusters (Fig. 6E, Suppl. Table 6). Analysis of the $\gamma\delta$ T1 TRAV13-4/DV7 CDR3 sequences revealed a high level of diversity that showed no evidence of clonal expansions (Fig. 6F). This was in contrast to the highly restricted V γ 6 repertoire in the c4 $\gamma\delta$ T17 cluster, as well as the restricted V γ 4 repertoire in the c2 $\gamma\delta$ T17 cluster that was dominated by semi-invariant TRDV5 and TRDV2-2 sequences (Fig. 6F). As no V δ 7-specific antibody was available, we independently confirmed this observation using single-cell PCR to assess the lung V γ 4 CDR3 δ sequences from 13 individual mice. These data demonstrated that, other than TRDV2-2 and TRDV5, TRDV7 is the only other TCR delta chain used at an appreciable frequency by V γ 4 T cells in the B6 mouse lung and that it exhibits significant diversity in its CDR3 (Suppl. Fig. 9). V γ 1 $\gamma\delta$ T1 cells, in contrast, exhibited a more mild pairing preference where 44% of the $\gamma\delta$ T1 clonotypes utilized TRAV15N-1, TRAV15-1/DV6-1, or TRAV15D-2/DV6D-2 and 13% utilized TRAV13-4/DV7 (Suppl. Table 6). Together these data formed a comprehensive map of B6 lung $\gamma\delta$ T cell transcriptomes and revealed that lung innate-like $\gamma\delta$ T1, like $\gamma\delta$ T17 and $\gamma\delta$ NKT, are all characterized by preferential TCR γ - and δ -chain pairings. This was especially notable for V γ 4 and our data suggested that V γ 4 $\gamma\delta$ T1 exhibited a distinct preference for the utilization of TRAV13-4(DV7) chains with diverse CDR3.

SAP regulates peripheral V γ 4/V δ 7 $\gamma\delta$ T1 cells and shapes the lung $\gamma\delta$ T17 repertoire

Next, we assessed the effect of SAP deficiency on the lung $\gamma\delta$ TCR repertoire by comparing B6 and B6.SAP^{-/-} lung $\gamma\delta$ T cells (Fig. 7A). Consistent with our previous observation that lung $\gamma\delta$ T cell IFN- γ production is decreased in SAP-deficient mice¹⁸, we observed a decrease in the frequency of the $\gamma\delta$ T1 c2, c3, and c7 clusters (Fig. 7B). Moreover, we observed a striking decrease in the frequency of TRGV4/TRDV7-expressing cells in the lungs of B6.SAP^{-/-} mice (Figs. 7C-D, Suppl. Table 6), and we noted that we found no evidence of a compensatory increase in the frequency of other TRDV-expressing lung V γ 4 $\gamma\delta$ T1 cells in the B6.SAP^{-/-} lung. We independently confirmed this SAP-dependent decrease in TRAV13-4/TRDV7 in both lung and spleen using qPCR (Suppl. Fig. 10), and we demonstrated a significant decrease in the number of CD44⁺SLAMF7⁺ V γ 4 T cells in both the lungs and spleen of B6.SAP^{-/-} mice (Fig. 7E, Suppl. Fig. 10). The decreased lung $\gamma\delta$ T1 did not appear to be due to a SAP-dependent decrease in homeostatic proliferation, nor to increased apoptosis, since we observed no significant change in lung $\gamma\delta$ T cell BrdU incorporation or Annexin V staining in B6.SAP^{-/-} mice (Suppl. Fig. 10).

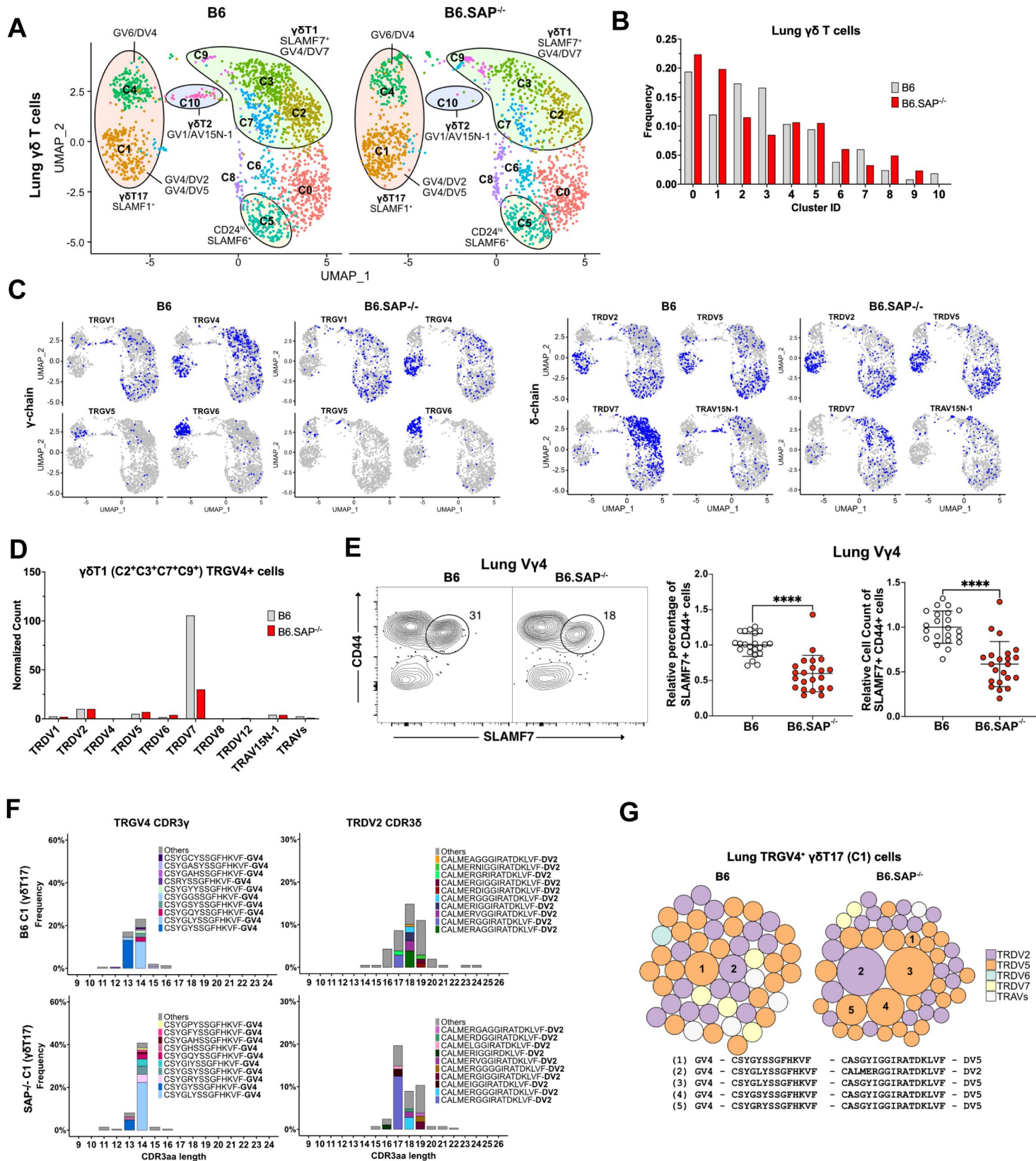


Figure 7.

Specific decrease in V γ 4/V δ 7 $\gamma\delta$ T1 cells in the lungs of SAP-deficient mice.

(A) UMAP representation of B6 (left) and B6.SAP^{-/-} (right) lung $\gamma\delta$ T cells, pooled from 3 mice per strain. **(B)** SAP-dependent variation in lung $\gamma\delta$ T cell cluster frequencies. The frequency of each B6 and B6.SAP^{-/-} cluster as a percentage of all lung $\gamma\delta$ T cells is shown. **(C)** UMAP representation of selected TRG and TRD chain V-segment usage (blue) among B6 and B6.SAP^{-/-} lung $\gamma\delta$ T cells. **(D)** SAP-dependent decrease in lung $\gamma\delta$ T1 TRGV4/TRDV7. Normalized TRDV counts in B6 and B6.SAP^{-/-} lung TRGV4⁺ $\gamma\delta$ T1 clusters (C2/C3/C7/C9). **(E)** SAP-dependent decrease in SLAMF7⁺ lung V γ 4 T cells. Representative contour plots of CD44 and SLAMF7 expression in B6 (left) and B6.SAP^{-/-} (right) lung V γ 4 $\gamma\delta$ T cells are shown at *left*. Relative frequencies and counts of CD44⁺SLAMF7⁺ lung V γ 4 cells is shown at *right*. Data represent the cumulative data from five independent experiments, ****p < 0.0001 using unpaired t-test. **(F)** SAP-dependent skewing of the lung $\gamma\delta$ T17 TCR repertoire. Amino acid length distributions of B6 and B6.SAP^{-/-} TRGV4⁺ CDR3 γ (left) and TRDV2⁺ CDR3 δ (right) sequences in the lung V γ 4 $\gamma\delta$ T17 C1 cluster. The top 10 clonotypes are color-coded and all other clonotypes are shown in gray. Bars represent the frequency of $\gamma\delta$ T cells in the c1 cluster. **(G)** TCR clonotype bubble plots depicting the top 50 lung TRGV4 clonotypes in the C1 $\gamma\delta$ T17 cluster. Bubble size indicates clonotype frequency as a percentage of C1 V γ 4 (n = 62 B6, 104 B6.SAP^{-/-}) T cells, and colors correspond to specific TRDV chains. Selected clonotypes are numbered and their corresponding CDR3 γ and CDR3 δ sequences are displayed below.

In addition to the decrease in lung $\gamma\delta$ T1, we also noted a loss of the c10 $\gamma\delta$ T2 cluster consistent with the enrichment in SAP-dependent $\gamma\delta$ NKT TCRs in this cluster, and we observed an increased frequency of the V γ 4 $\gamma\delta$ T17 c1 cluster (**Fig. 7B**). Given our previous observation of a SAP-dependent decrease in lung $\gamma\delta$ T cell IL-17 production in young, but not adult mice¹⁸, and our findings here of SAP-dependent alterations in $\gamma\delta$ T17 clonotypes, we were interested in determining whether the adult lung $\gamma\delta$ T17 repertoire was altered in SAP-deficient mice. Indeed, a comparison of the lung V γ 4 c1 $\gamma\delta$ T17 CDR3 sequences between B6 and B6.SAP^{-/-} mice revealed significant alterations in the $\gamma\delta$ T17 c1 cluster TCR repertoire. Specifically, we noted a decreased frequency of TRGV4 chains that exhibited a germline-encoded CDR3 γ (CSYGYSS) and an increased frequency of TRGV4 chains with a CDR3 γ exhibiting evidence of N/P additions (CSYG(X)YSS). These changes were associated with a significant shift in the usage of TRDV2-2 CDR3 δ sequence usage (**Fig. 7F**) and an altered c1 TCR clonotype frequency in B6.SAP^{-/-} mice (**Fig. 7G**). Although the number of germline-encoded invariant TRGV4/TRDV5 clone (clonotype 1 in **Fig. 6B**) was diminished in B6.SAP^{-/-} lung $\gamma\delta$ T cells (**Fig. 7G**, **Suppl. Table 6**), the low numbers of cells available for analysis precluded us from making strong conclusions regarding the loss of any one particular clonotype. Taken together, these data suggest that V γ 4 $\gamma\delta$ T1 preferentially utilize the TRAV13-4(DV7) chain in the lung and that the SAP-dependent decrease in peripheral $\gamma\delta$ T cell IFN- γ production is not due to dysregulated cytokine production, but rather due to a specific loss of innate-like $\gamma\delta$ T1 subsets. This finding, together with our finding that the adult lung $\gamma\delta$ T17 TCR repertoire is altered in SAP-deficient mice suggests that SLAM/SAP signaling plays a role in shaping the $\gamma\delta$ TCR repertoire.

Discussion

A common denominator in most innate-like T cell developmental programming is a dependence on the SLAM/SAP signaling pathway^{11, 18, 19, 54}. In the $\gamma\delta$ T cell lineage, SAP plays a critical role in the development of IFN- γ and IL-4-producing $\gamma\delta$ NKT cells²¹, as well as in $\gamma\delta$ T1 and $\gamma\delta$ T17¹⁸, but exactly when during $\gamma\delta$ T cell development SAP functions, and how it exerts these effects is unknown. Here, we provide evidence that SAP mediates its effects on immature, uncommitted thymic $\gamma\delta$ T cells and that it works to simultaneously promote progression through

the $\gamma\delta$ T17 developmental pathway while inhibiting the diversion of $\gamma\delta$ T cells into the $\alpha\beta$ T cell pathway. The redirection of $\gamma\delta$ T cell clonotypes into the $\alpha\beta$ T cell pathway at E17 is associated with a decreased frequency of these mature clonotypes in neonatal thymus, and an altered $\gamma\delta$ T17 TCR repertoire in the periphery. Finally, we identify V γ 4/V δ 7 T cells as a novel, SAP-dependent IFN- γ -producing V γ 4 $\gamma\delta$ T1 subset.

The developmental pathway(s) that gives rise to $\gamma\delta$ T cell functional subsets remains unclear. $\gamma\delta$ T cells develop at the DN2/DN3 stage after receiving a sufficiently strong $\gamma\delta$ TCR-mediated signal⁵⁵. Within the continuum of signals that can drive $\gamma\delta$ T cell development, it is the relatively weaker TCR signals that promote $\gamma\delta$ T17 programming while the stronger TCR signals foster $\gamma\delta$ NKT and $\gamma\delta$ T1 programming^{4, 8, 11, 14, 56, 57}. Despite considerable effort, the precise thymic developmental stages at which $\gamma\delta$ T cell lineage diversification occurs remains unclear. Recent studies examining this issue have demonstrated that effector programming can occur at a CD44^{neg}CD45RB^{neg} stage^{4, 14}, and that c-Maf-directed $\gamma\delta$ T17 programming occurs at a CD45RB^{low} stage³⁶. In addition, by altering TCR signal strength in a transgenic model, Chen et al., defined a *Ccr9*-enriched cluster as a point of effector lineage diversification¹⁰. Our findings here suggest that SAP influences $\gamma\delta$ T17 developmental programming during the immature uncommitted CD24^{hi}CD73^{neg} stage, either during or soon after *Il2ra* (CD25) downregulation. The SAP-dependent c2 cluster we define here was highly enriched in numerous genes (e.g., *Sox13*⁴⁰, *Blk*³⁵, *Maf*³⁶, *Etv5*³⁹, *Sh2d1a*¹⁸) previously associated with $\gamma\delta$ T17 development as well as *Ccr9*, and the cell surface phenotype of c2 $\gamma\delta$ T cells is SLAMF1^{pos}SLAMF6^{pos}CD44^{low/high}CD45RB^{low}. Altogether, these data suggest that the SAP-dependent c2 cluster is identical to these previously defined developmental checkpoints. Given that these previous studies have demonstrated that this developmental checkpoint is influenced by TCR signaling, it is interesting to note that SLAM/SAP signaling has previously been demonstrated to foster semi-invariant NKT cell differentiation through its ability to regulate TCR signal strength⁵⁸.

To that end, we noted that in addition to a decrease in the number of immature $\gamma\delta$ T cells expressing BLK, the expression level of BLK was decreased in B6.SAP^{-/-} $\gamma\delta$ T cells. BLK is a member of the Src tyrosine kinase family whose members also include LCK and FYN. All three of these kinases have been implicated in $\gamma\delta$ T cell development⁵⁹ and BLK has been shown to promote $\gamma\delta$ T17 development³⁵. Thus, it's possible that decreased BLK expression could contribute to the impaired $\gamma\delta$ T17 development observed in SAP-deficient mice. In addition, since SAP has been shown to regulate conventional $\alpha\beta$ T cell differentiation through its ability to interact with FYN⁶⁰, it's possible that SAP interacts with one or all of these Src kinase family members to exert its effects. Whether SAP works to regulate TCR signal strength during $\gamma\delta$ T cell development, or whether it exerts its effects through a TCR-independent mechanism is currently under investigation.

Nevertheless, our finding that, in addition to regulating the frequency of the c2 cluster, SAP inhibits the number of $\gamma\delta$ T cells that enter the $\alpha\beta$ T cell developmental pathway, is reminiscent of previous work using transgenic TCR models demonstrating that a reduction of TCR signal strength results in a redirection of $\gamma\delta$ T cells to the $\alpha\beta$ T cell lineage^{53, 61}. In addition, Fahl et al. recently demonstrated that both entry into the $\gamma\delta$ T cell lineage as well as $\gamma\delta$ T cell effector programming involved repression of TCF1 expression by TCR-driven increases in *Id3*⁴². Consistent with these results, we found that *Tcf7* (encoding TCF1) transcript levels were highest in the $\alpha\beta$ T cell-like c6 cluster, and we note that *Tcf7* transcripts levels were noticeably higher in the *Il2ra*-enriched post-selection c5 cluster in B6.SAP^{-/-} $\gamma\delta$ T cells. *Id3* expression, however, did not appear to be altered in SAP-deficient mice. Whether SAP-mediated signals could alter $\gamma\delta$ T cell *Tcf7* expression through a TCR-independent mechanism remains an open question. Indeed, recent evidence that some DN1 thymocytes possess a SOX13^{hi} IL-17-like phenotype and that V γ 4 $\gamma\delta$ T17 arise from DN1

precursors^{12, 62} indicates the existence of such a TCR-independent developmental programming stage. It may be possible, therefore that SLAM/SAP-mediated signals could regulate the number and/or fitness of these DN1 precursors prior to the expression of TCR.

While our data indicate that SAP regulates the diversion of at least some TCR clonotypes into the $\alpha\beta$ TCR pathway, the fate of those cells is unclear. Certainly, the fact that the clonotypes that were diverted to the $\alpha\beta$ T cell pathway in E17 thymus are under-represented in the mature D9 thymus $\gamma\delta$ T17 clusters suggests that this diversion has significant consequences on the $\gamma\delta$ T cell receptor repertoire, but whether these diverted $\gamma\delta$ T cells contribute to the $\alpha\beta$ T cell compartment is unclear. Recently, a population of hybrid $\alpha\beta$ - $\gamma\delta$ T cells was described in both mouse and human⁶³ that responds to peptide/MHC stimulation like conventional $\alpha\beta$ T cells, as well as to IL-1 and IL-23 which induce both IFN- γ and IL-17 production⁶³. In the mouse, at least some of these cells are V γ 4/V δ 5, V γ 4/V δ 4, or V γ 4V δ 7 and are present in the embryonic thymus, raising the possibility that $\gamma\delta$ T cells diverted to the $\alpha\beta$ T cell pathway in B6.SAP^{-/-} mice contribute to this $\alpha\beta$ - $\gamma\delta$ T cell population. While we have not observed a significant increase in the overall numbers of $\alpha\beta$ - $\gamma\delta$ T cells in B6.SAP^{-/-} thymus (not shown), a direct comparison of B6 and B6.SAP^{-/-} $\alpha\beta$ - $\gamma\delta$ TCRs may be necessary to see such an effect.

Our previous observations that SAP deletion did not completely abolish the development of IL-17- and IFN- γ producing $\gamma\delta$ T cells¹⁸ suggested the possibility that SAP may be critical for the development of additional $\gamma\delta$ T cell clonotypes other than the V γ 1V δ 6.3 subset. Therefore, we compared the SAP-dependent and SAP-independent $\gamma\delta$ TCR repertoires using paired TCR clonotype analysis, which allowed us to track clonotype fate during development. Our findings not only confirmed previous results demonstrating highly restricted $\gamma\delta$ T17^{45, 46}, and $\gamma\delta$ NKT²¹ TCR repertoires, they also revealed that the vast majority of V γ 4 $\gamma\delta$ T1 subsets in the periphery preferentially utilize TRGV4/TRAV13-4(DV7) clonotypes, thereby further highlighting the strong links between $\gamma\delta$ T cell function and the utilization of specific TCRs²³. The reason for this preferential TRDV usage in V γ 4/V δ 7 $\gamma\delta$ T1 is unclear. While the TRAV13-4(DV7) CDR3 sequences were diverse and did not suggest evidence of clonal expansion, it may be possible that TRDV-encoded sequence motifs could be required to interact with a ligand, similar to the way that TRGV-encoded sequences mediate recognition of butyrophilins^{64, 65}.

Exactly when during development SAP exerts its effects on V γ 4/V δ 7 T cells is still somewhat unclear. We found comparatively low numbers of TRGV4/TRDV7 $\gamma\delta$ T cells in embryonic, neonatal, or adult thymus, and those that were identified were generally found in the immature clusters. The increased frequency in B6.SAP^{-/-} E17 thymus of TRGV4/TRDV7 $\gamma\delta$ T cells in the *Il2ra*⁺ c5 cluster coupled with their decreased frequency in the emigrating *S1pr1*⁺ c4 cluster suggests a model where SAP promotes progression through the early stages of development through thymic egress. SAP deficiency, therefore, could result in lower V γ 4/V δ 7 output. However, this model is at odds with our finding that the frequency of *S1pr1*⁺ clusters in neonatal thymus is increased in B6.SAP^{-/-}, and that these clusters contained increased numbers of TRGV4/TRDV7 $\gamma\delta$ T cells. A possible explanation could be found in a previous report demonstrating that a V γ 4/V δ 7 subset increases in frequency in the periphery of B6 mice over time and undergoes extrathymic expansion⁶⁶. Our finding here that the V γ 4/V δ 7 T cells represent a SAP-dependent innate-like $\gamma\delta$ T1 subset raise the possibility that SAP regulates the homeostatic expansion of this subset after it leaves the thymus. The mechanism, however, through which SAP is regulating the number of the V γ 4/V δ 7 subset remains unclear as we have so far observed no evidence of changes in homeostatic proliferation or apoptosis of these cells in the lungs.

We previously demonstrated that the SAP-dependent dependent impairment in $\gamma\delta$ T17 function in the neonatal periphery largely corrects itself as mice mature¹⁸. This suggested the possibility that SAP-dependent $\gamma\delta$ T17 clonotypes originating in the embryonic thymus were being replaced over time by SAP-independent $\gamma\delta$ T17 clonotypes. Although our data certainly point to a SAP-dependent decrease in a germline-encoded V γ 4/V δ 5 clonotype in neonatal thymus, the low

numbers of this clonotype in the lung makes it difficult to draw definitive conclusions in the periphery. Still, we did observe a role for SAP in shaping the peripheral $\gamma\delta$ T17 repertoire, as we noted we noted a decreased frequency of germline-encoded TRGV4 sequences (so-called “GYSS”⁶⁷) coupled with an increased frequency of TRGV4 sequences with more diverse CDR3 (“GXYSS”) in the periphery. Although the potential significance of this change on $\gamma\delta$ T cell function is unclear, it does support a model in which SAP regulates $\gamma\delta$ T17 developmental programming during embryonic thymic development, where the low expression of terminal deoxynucleotide transferase results in a high frequency of germline-encoded V-J junctions.

In summary, our findings reveal the presence of multiple SAP-dependent developmental checkpoint during the very early stages of $\gamma\delta$ T cell development in the thymus, indicating a more complex role for this signaling pathway in $\gamma\delta$ T cell developmental programming than previously appreciated. Indeed, our data indicate that SAP functions to regulate entry into the $\gamma\delta$ and $\alpha\beta$ T cell developmental pathways, and we make the striking observation that nearly all V γ 4 $\gamma\delta$ T1 cells in the periphery preferentially use V87 chains and that it is a decrease in the V γ 4/V87 subset that, in part, underlies the SAP-dependent decrease in $\gamma\delta$ T cell IFN- γ production. Whether SAP regulates these phenotypes through regulation of TCR signal strength or additional pathways remains an open question. Taken together, the data presented here suggest a key role for the SLAM/SAP signaling pathway in generating and/or maintaining the innate-like phenotype in the $\gamma\delta$ T cell lineage.

Methods

Mice

C57BL/6J (B6) and B6.*Sh2d1a*^{-/-} (SAP^{-/-}) mice were purchased from the Jackson Laboratory (Bar Harbor, ME) and were housed and bred in a specific pathogen-free facility at the animal facility of the University of Vermont. For experiments using E17 mice, timed-pregnant matings were used. The presence of a vaginal plug indicated day 0 of pregnancy. The flow cytometry experiments included 8-13 weeks old B6 and B6.SAP^{-/-} mice (age and sex-matched). All experimental procedures on animals were carried out with the approval of the University of Vermont Institutional Animal Care and Use Committee.

Tissue Processing

Single-cell suspensions from the spleen and thymus were obtained by gently passing tissue through nylon mesh. Red blood cells (RBCs) were lysed with Gey’s solution. Single-cell suspensions from the lungs were prepared as described (Benoit et al., 2015). Briefly, dissected lungs were finely minced in a digestion buffer containing DMEM, 1 mg/ml collagenase type IV (Gibco), and 0.2 mg/ml DNase (Sigma-Aldrich). After shaking at 200 rpm at 37°C for 20 min, the lung digests were triturated with a 16-gauge needle, followed by an additional 20 min shaking at 200 rpm at 37°C. Following the final trituration, red blood cells (RBC) were lysed with Gey’s solution. Finally, the digested lung tissues were filtered through a 70- μ m filter and washed in PBS/2%FBS. Cells were counted either on a MACSQuant VYB or using a hemacytometer following trypan blue staining.

Flow Cytometry

Cells were incubated with LIVE/DEAD™ Fixable Blue Dead Cell stain (Thermo Fisher Scientific, Grand Island, NY) for 30 min at 4°C in PBS, after which they were resuspended in Fc-Block (BioLegend) containing PBS/2% FCS/0.1% sodium azide buffer. After a 5-10 min incubation on ice, antibodies specific for cell surface markers were added, incubated for 30 min on ice, and washed in PBS/2% FCS/0.1% sodium azide buffer. Brilliant Violet Staining buffer (BD Biosciences) was used in experiments contain more than one Brilliant Violet, Brilliant UV, or SuperBright-conjugated Ab. Cells were fixed in PBS/1% PFA buffer and stored at 4° C prior to data collection.

For nuclear staining, cells were permeabilized with Foxp3 transcription factor fixation/permeabilization buffer (ThermoFisher) following surface staining. Cells were incubated overnight at 4°C, washed, and incubated with 25 micrograms/ml total rat IgG before adding Abs against transcription factors. After staining, cells were washed and resuspended in PBS/2% FCS/0.1% sodium azide buffer. For intracellular cytokine staining, cells were permeabilized with PBS/1% FCS/0.1% sodium azide/0.1% saponin following surface staining and fixation. To reduce unspecific binding, 25 µg/ml total rat IgG was added prior to cytokine staining with fluorescent-labeled antibodies.

Flow cytometry data were collected on a Cytex® Aurora (Cytex Biosciences) spectral cytometer, and data analysis was performed using FlowJo Software (Tree Star). In some cases, visualization of flow cytometry data was conducted by concatenation of individual .fcs files, followed by dimensionality reduction and clustering using UMAP and FlowSOM⁶⁸ in FlowJo. Antibodies used in these experiments are anti-CD4 (RM4-5), anti-CD11b (M1/70), anti-CD11c (N418), anti-CD19 (1D3), anti-CD24 (M1/69), anti-CD25 (PC61), anti-CD27 (LG.3A10), anti-CD44 (IM7), anti-CD45RB (C363-16A), anti-CD73 (TY/11.8), anti-CD150 (TC15-12F12.2), anti-CD196 (29-2L17), anti-CD319 (4G2), anti-cMAF (sym0F1), anti-IFNγ (XMG1.2), anti-IL17a (TC11-18H10.1), anti-IL4 (11B11), anti-PLZF (9E12), anti-T-bet (4B10), anti-TCRβ (H57-597), anti-Vγ5 (536), and anti-Vd6.3 (C504.17c) from BioLegend; and anti-RORγt (Q31-378), anti-TCR8 (GL3), anti-Vγ1 (2.11) from BD biosciences; and anti-CD8a (53-6.7), anti-CD45 (30-F11), anti-CD200 (OX90), anti-CD278 (7E17G9), anti-Ki-67 (SolA15), anti-Ly108(13G3-19D), anti-Vγ4 (UC3-10A6) from eBiosciences; anti-S1P1 (713412) from R&D systems; anti-Blk (polyclonal) from Cell Signaling Technology; and anti-17D1

In vivo proliferation and apoptosis

For *in vivo* proliferation analysis, each mouse was intraperitoneally injected with 1 mg of 5-bromo-deoxyuridine (BrdU) (Sigma-Aldrich) each day for 3 days. Lung cell suspensions were prepared as above, stained with surface markers, after which they staining with anti-BrdU antibody according to the manufacturer's instructions (BrdU Flow Kit; BD Biosciences). For apoptotic cell detection, annexin V staining was performed. Briefly, surface staining was followed by dead cell staining with Fixable Viability Dye eFluor™ 780 dye. The cells were washed and resuspended in an annexin-binding buffer (10 mM HEPES, 140 mM NaCl, and 2.5 mM CaCl₂, pH 7.4). Finally, 1 µl of Annexin V Conjugate (Thermo Fisher Scientific, Grand Island, NY) was added to each tube, and cells were incubated for 15 minutes at room temperature (RT) prior to data collection.

Bulk RNA Sequencing and data analysis

For the bulk RNA sequencing, thymocytes from 10-day old B6 (n=4 male, n=2 female) and B6.SAP^{-/-} (n=5 male, n=1 female) mice were prepared by depleting CD4⁺ cells from thymocytes using the EasySep™ Mouse CD4 Positive Selection Kit II (STEMCELL Technologies). Cells were then incubated with Fixable Viability Dye eFluor™ 780 (Thermo Fisher Scientific, Grand Island, NY) for 30 min at 4°C in PBS, after which they were resuspended in Fc-Block (BioLegend) containing PBS/2% FCS. After washing, cells were stained with antibodies against CD11b (M1/70), CD19 (1D3), CD24 (M1/67), CD45 (30-F11), TCRβ (H57-597), and TCR8 (GL3) for 30 min at 4°C. Following surface staining, CD24^{hi}TCR8⁺ cells were sorted from a CD11b⁻CD19⁻TCRb⁻population of live CD45⁺ lymphocytes. Immature (CD24^{high}) thymocytes from both B6 and B6.SAP^{-/-} mice were sorted directly into 350 microliters of RLT lysis buffer (Qiagen). RNA was extracted using the RNeasy Micro kit (Qiagen) according to manufacturer instructions. After RNA quantification and quality assessment, cDNA libraries were prepared using SMARTer Stranded Total RNA-Seq Kit v3 - Pico Input Mammalian (Takara, Japan) according to the manufacturer's protocol. Following library clean-up and quantification, the libraries were sequenced (single-end 75 bp) on an Illumina HiSeq1500.

The Fastq files were checked for data quality using FastQC and MultiQC⁶⁹ bioinformatics tools. Cutadapt⁷⁰ was used to filter out low-quality sequences and to remove adapter sequences from all reads. The expression of transcripts was quantified using the mapping-based mode of Salmon⁷¹, which quasi-mapped RNA-seq reads to an index created from a mouse reference transcriptome (GRCm39). The transcript abundance estimates were imported using tximport⁷² and differential gene expression (DE) analysis was performed using DESeq2⁷³ employing the Wald test of the negative binomial model coefficients to determine statistical significance. The results were analyzed based on Log₂ fold change (Log₂FC ≥0.5 or ≤-0.5) and adjusted P-value (padj ≤0.0001) to determine whether a gene was differentially expressed.

Cell sorting and library preparation for CITE-seq and V(D)J enrichment

For thymus single-cell transcriptome analysis, we used embryonic day 17 (E17) B6 (n=4) and B6.*Sh2d1a*^{-/-} (SAP^{-/-}) (n=4) mice, nine day-old (D9) B6 (n=3) and B6.SAP^{-/-} (n=3) mice, and six-week-old B6 (n=3 male) and B6.SAP^{-/-} (n=2 male, n=1 female) mice. For single-cell analysis of adult lung γδ T cells, we used 13 week old B6 and B6.SAP^{-/-} littermates (pooled cells from 2 males and one female mouse in each group).

Analysis of gene expression, surface protein expression, and TCR V(D)J repertoire was performed using the 10X Genomics Chromium Next GEM Single Cell 5' V(D)J with feature barcoding kit (version 1.1). To sort cells for scRNAseq, thymocytes were stained with anti-TCRδ-PE (GL3) antibody and enriched using anti-PE MicroBeads kit (Miltenyi Biotec). The exception was E17 thymus which did not undergo TCRδ enrichment. Following live/dead staining with Fixable Viability Dye eFluor™ 780 (Thermo Fisher Scientific, Grand Island, NY) for 30 min at 4° C in PBS, cells were resuspended in Fc-Block (BioLegend) containing PBS/2% FCS. Cells were then simultaneously stained with oligonucleotide-conjugated antibodies targeting CD24, CD44, CD45RB, CD73, SLAMF1, SLAMF6, Vγ1, and Vγ4 (TotalSeqC-BioLegend), oligonucleotide-conjugated hashtag antibodies (TotalSeqC-BioLegend), and fluorophore-conjugated antibodies for cell sorting which included the anti-CD11b (M1/70), anti-CD11c (N418), anti-CD19 (1D3), anti-CD45.2 (104) and anti-TCRβ (H57-597) antibodies. Following surface staining of the enriched population, TCRδ⁺ cells were sorted from a CD11b⁺CD11c⁺CD19⁺TCRβ⁺ population of CD45.2⁺ live lymphocytes.

Libraries were prepared according to the manufacturer's instructions with modifications to the V(D)J enrichment procedure. Equal numbers of live GL3⁺ γδ T cells from B6 and B6.SAP^{-/-} mice were pooled and captured (between 5000 – 7000 cells) using a 10x Chromium controller. Following cDNA preparation, the gene expression and CSP libraries were created according to the manufacturer's instructions. To obtain TCR gamma and delta chain sequences, we replaced kit-provided enrichment primers with custom mouse γ and δ chain-specific primers and modified the number of PCR cycles used from manufacturer recommended cycling conditions. In the first enrichment step, an equal amount of cDNA was used as a template for PCR amplification of γ and δ chains separately using a total of 12 and 10 PCR cycles, respectively. The primers used for the first γ-chain enrichment are: Forward Primer 5'-AATGAT ACGGCGA CCACCG AG ATCTACTCTT TCCCTACAC GACGCTC-3' (2 μM) and Reverse Outer Primers 5'-GGAA AGAAGT TTCAAGGAS ACAAAG-3' (1 μM), 5'-CCCTTATG ACTTCAGG AAAGAA CTTT-3' (0.5 μM). The primers used for the first δ-chain enrichment are: Forward Primer 5'- AATGAT ACGGCGACCACCG AGATCTACTCT TTTCCCTAC ACGACGCTC-3' (2 μM) and Reverse Outer Primer 5'-CCACAATC TTCTTGGATG ATCTGAG-3' (0.5 μM). Following PCR product purification, the γ-chain and δ-chain amplicons went through the second round of enrichment separately using a total of 12 and 10 PCR cycles, respectively. The primers used for the second γ-chain enrichment are: Forward Primer 5'-AATGA TACGGCGACCA CCGAGATCT-3' (0.5 μM) and Reverse Inner Primers 5'-ACAAA GGTATGTCCA GTCTTATGGA-3' (0.5 μM), 5'-GGAGACA AAGGTAGGT CCCAGC-3' (0.5 μM). The primers used for the second δ-chain enrichment are: Forward Primer 5'-AATGATA CGGCGACCACC GAGATCT-3' (0.5 μM) and Reverse Inner Primer 5'-GTCACCTC TTTAGGGTAG AAATCTT-3'. Following PCR product cleanup, the second

round PCR products (γ -chain and δ -chain amplicons) were quantified and combined in equal quantity making the final TCR V(D)J library. Finally, all three libraries (gene expression, cell surface protein expression, and TCR V(D)J) were sequenced (paired-end 26 x 91 bp) on an Illumina HiSeq 1500 sequencer or a NextSeq 2000.

CITE-seq quality control, filtering and demultiplexing

Raw sequence reads from the gene expression and CSP libraries were processed using the Cell Ranger software (v4.0.0)⁷⁴. Reads were mapped to the GRCm38 (mm10) mouse reference genome. To exclude the possibility of TCR transcripts playing a role in UMAP clustering, all TCR genes were omitted from the analysis using the “cellranger reanalyze” argument of Cell Ranger software.

Quality control, filtering, UMAP projection, and clustering analysis were conducted using Seurat (v4.3.0.1)⁷⁵ R package. Briefly, cells with a low number (≤ 2000 for E17 thymus dataset, ≤ 1200 for D9 thymus dataset and ≤ 750 for lung dataset) of unique genes (nFeature_RNA), a frequency of mitochondrial reads $> 4\%$ and a frequency of ribosomal protein $< \sim 12\%$ were eliminated from further analysis. To reduce cell cycle-related and stress-related heterogeneity in the normalized data, selected cell cycle genes and stress-induced genes⁷⁶ were regressed out during the data normalization step. Note that in some instances, contaminating clusters expressing macrophage specific marker genes were manually removed using Seurat “CellSelector ()” argument. We used the Seurat function “HTODemux()” to demultiplex single cells back to their sample of origin⁷⁷.

CITE-Seq and V(D)J enrichment data analysis

The demultiplexed datasets were further analyzed for gene expression, cell surface protein expression following the multimodal analysis pipeline of Seurat⁷⁵. For each dataset, we performed normalization and variance stabilization of the gene expression data using the scTransform⁷⁸. Following identification of top significant principal components (PCs) using the principal component analysis (PCA), we performed a graph-based (UMAP) semi-supervised clustering of cells with similar gene expression patterns using specific number of top PCs (E17 = 25, D9 = 30, W6 = 27, lung = 28) and resolution values (E17 = 1.2, D9 = 1.6, W6 = 0.65, lung = 1.2) for each dataset. The cell surface protein expression data (Centered Log Ratio (CLR) normalized) from CSP libraries as well as the gene expression data were then visualized on the clustered cells using the R package scCustomize⁷⁹. The differentially expressed features of each cluster (cluster biomarkers) were identified using FindAllMarkers() function using the Wilcoxon Rank Sum test. For identifying differentially expressed genes (DEGs) between B6 and B6.SAP^{-/-} samples, we performed a pseudobulk DE (differential expression) analysis of aggregated read counts using the DESeq2 package, which accounts for biological replicates making it a better approach than other specialized single-cell DE analysis methods⁸⁰. Briefly, for cells of a given sample type, we aggregated RNAseq reads across biological replicates identified by individual hashtags. Then transformed the genes-by-cell matrix to a genes-by-replicates matrix and performed DE analysis using the DESeq2⁷³ package employing the Wald test of the negative binomial model coefficients to determine statistical significance. Finally, the trajectory and pseudotime inference analysis was performed using the partition-based graph abstraction (PAGA)⁸¹ analysis, which is a part of the Scanpy⁸² python package.

For the analysis of V(D)J library sequences, Cell Ranger v6.1.2 (“cellranger vdj” argument) was used to map the TCR V(D)J reads to a custom-made reference containing sequences from the IMGJ database⁸³. The Cell Ranger V(D)J output was then processed with custom bashscripts. Briefly, the TRGV and TRAV/TRDV contigs identified by Cell Ranger V(D)J were demultiplexed to individual barcodes after which the dominant TRGV and TRAV/DV transcripts were selected per cell. Finally, the V(D)J information of each cell was integrated with scCITE-seq data and visualized using a combination of Seurat and custom-made R scripts. We note that since about half of Vy1, Vy4, and

Vy7 T cells contain functionally rearranged TRGV2 chains, but only a small percentage express Vy2 on the cell surface⁸⁴, the methods used here cannot adequately assess the true TRGV2 repertoire. Therefore, we report it here for accuracy but have omitted TRGV2 from our analysis.

Quantitative PCR

Lung qPCR samples were prepared by positively selecting leukocytes using CD45 MicroBeads (Miltenyi Biotec). Spleen qPCR samples were prepared by depletion of CD19⁺ TCRβ⁺ cells using biotinylated antibodies and Streptavidin MicroBeads (Miltenyi Biotec) according to the manufacturer's instructions. Total RNA was extracted from approximately 5-10 million cells from each organ using the RNeasy Mini Kit (QIAGEN). SuperScript® IV Reverse Transcriptase (Thermo Fisher Scientific, Grand Island, NY) was used to generate cDNA using 400 ng of RNA and oligodT primer according to the manufacturer's instructions. qPCR was performed using 1 μl of cDNA template in iTaq Universal SYBR Green Supermix (Bio-Rad) containing TCR delta chain specific primer pairs. The qPCR reaction was run on a Bio-Rad CFX96 qPCR machine using the following cycling conditions: 95°C for 30 sec; 40 cycles of [95°C for 5 sec; 60°C for 30 sec; 72°C for 30 sec]. The β2M and TRDC genes were used as endogenous housekeeping controls. Primers used were as follows: mb2mforward, 5'- CATGGCTCG CTCGGTGACC-3', mb2mreverse, 5'-AATGTGAGGC GGGTGGAACTG-3', TRDCforward, 5'- CTCCGGCCA AACCATCTGTT-3', mouseDV2, 5'-CCAAGAAG CATAAAGCAGT ATAATG-3', mouseDV5, 5'- CCCATGA TGCAGATTT TGTTCAGG-3', mouseDV7, 5'- GGAAG MCTCGTCAGC CTGTTGT-3', mouseDCrev, 5'- CCACAATCTT CTTGGAT GATCTGAG-3'. Some primer sequences were adopted from Wei. et. al. (2015)⁴⁵.

Vy4 single cell sorting and TCR Sequencing

The plate-based based γδ TCR repertoire analysis of single-cell sorted Vy4 cells involved adult lungs from 8-10 weeks old B6 male (n=7) and female mice (n=6). Lung Vy4 TCR sequences were obtained using a protocol modified from Wei *et al.*, (2015)⁴⁵. Briefly, lung cell suspensions were prepared as described above. After washing, cells were resuspended in Fc-Block (BioLegend) containing PBS/2% FCS buffer, and stained with anti-CD11b (M1/70), anti-CD11c (N418), anti-CD19 (1D3), anti-CD45 (30-F11), anti-TCRδ (GL3) and anti-Vy4 (UC3-10A6) from for 30 min at 4°C in PBS. Fixable Viability Dye eFluor™ 780 (Thermo Fisher Scientific, Grand Island, NY) was used to identify dead cells. Single-cell sorting of lung Vy4⁺TCRδ⁺ from a CD11b⁻ CD11c⁻CD19⁻TCRβ⁻ population of CD45⁺ live lymphocytes were performed using a FACS Aria III cell sorter (BD Bioscience). Cells were sorted directly into 12.5 μl (per well) of OneStep RT-PCR Buffer mix (QIAGEN) in a 96-well PCR plate (one cell/well).

TCR sequences were then amplified and barcoded according to (Wei *et al.*, 2015). Briefly, the first two nested PCR reactions involved primer-specific amplification of gamma and delta TCR sequences. In both the first and second PCR reactions, the final concentration of each V-region primer is 0.36 μM, and each C-region primer is 0.6 μM. The first RT-PCR step was performed using the following cycling conditions: 50°C for 30 min; 95°C for 5 min; 25 cycles of [94°C for 30 sec; 62°C for 1 min; 72°C for 1 min]; 72°C for 7 min; 4°C. Next, 1 μl of the first round PCR product was used as a template for the second nested PCR reaction using HotStartTaq Master Mix Kit (QIAGEN). The cycling conditions for the second PCR reaction were 95°C for 5 min; 25 cycles of [94°C for 30 sec; 64°C for 1 min; 72°C for 1 min]; 72°C for 7 min; 4°C. In the third PCR reaction, 1 μl of second round PCR product as the template in a new 14 μl reaction using HotStartTaq Master Mix Kit (QIAGEN). Each well was barcoded. The final concentration of 5' and 3' barcoding primers was 0.05 μM each and paired-end primers were 0.5 μM each. The cycling conditions for the third PCR reaction were 95°C for 5 min; 36 cycles of [94°C for 30 sec; 62°C for 1 min; 72°C for 1 min]; 72°C for 7 min; 4°C. The γ and δ chain products were run on 1.2% agarose gel separately and purified using the Qiagen QIAquick Gel Extraction Kit according to kit instructions. The final products were then analyzed using the 2100 Bioanalyzer (Agilent Technologies). Finally, the γ and δ TCR products were mixed in equal concentration and sequenced on the Illumina MiSeq platform using the Illumina MiSeq Reagent Kit v2 (500-cycles).

The paired-end TCR sequencing reads were analyzed using a custom software pipeline capable of de-multiplexing barcoded sequences to individual cells. The resulting de-multiplexed reads were then analyzed by MiGMAP software, which identifies CDR3 sequences using the IgBlast 1.4.0 (NCBI) V(D)J mapping tool⁸⁵,⁸⁶. The single-cell TCR V(D)J profiling data were then visualized using custom R scripts.

Statistical analysis

Prism (GraphPad Software, San Diego, CA) was used for the statistical analysis of data. Unpaired Student t-tests, one-way ANOVA, or two-way ANOVA were used where appropriate. Tests were considered significant when p-values were less than 0.05. Error bars represent standard deviation (SD) unless otherwise indicated.

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Author contributions

S.K.M. and J.E.B. conceived and designed the project; S.K.M., J.E.B., E.A., K.J.H., B.M.H., R.S., and O.D. performed experiments and analyzed data; K.J.H., D.G., J.T.R., and N.S. aided with the single-cell TCR sequencing; D.M. assisted with bulk RNA sequencing; S.K.M. and J.E.B. performed the bioinformatics analysis of bulk RNA-seq and single-cell CITEseq (with immune profiling) data; J.E.B. supervised the project; S.K.M. and J.E.B. prepared the manuscript.

Competing interests

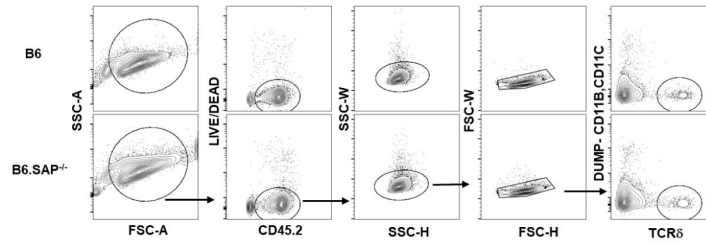
All authors declare no competing interests.

Materials and Correspondence

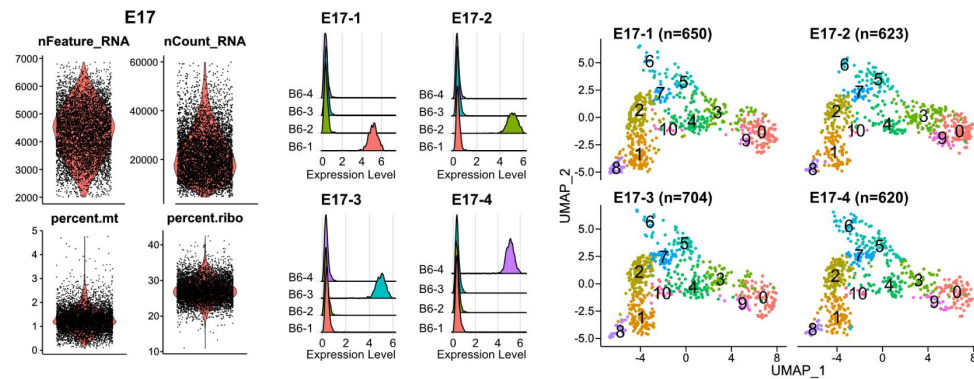
Correspondence and requests for materials should be addressed to J.E.B. (email: jonathan.boyson@med.uvm.edu)

Supplementary Figure Legends

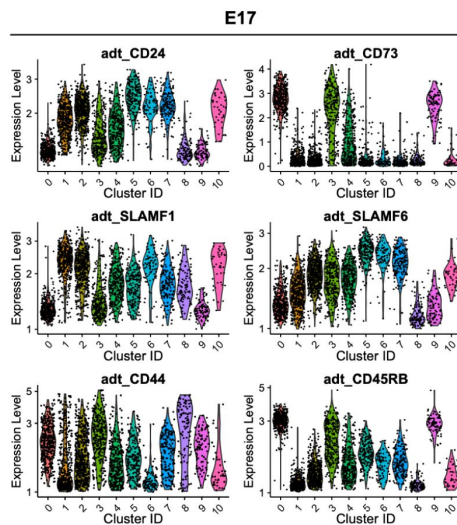
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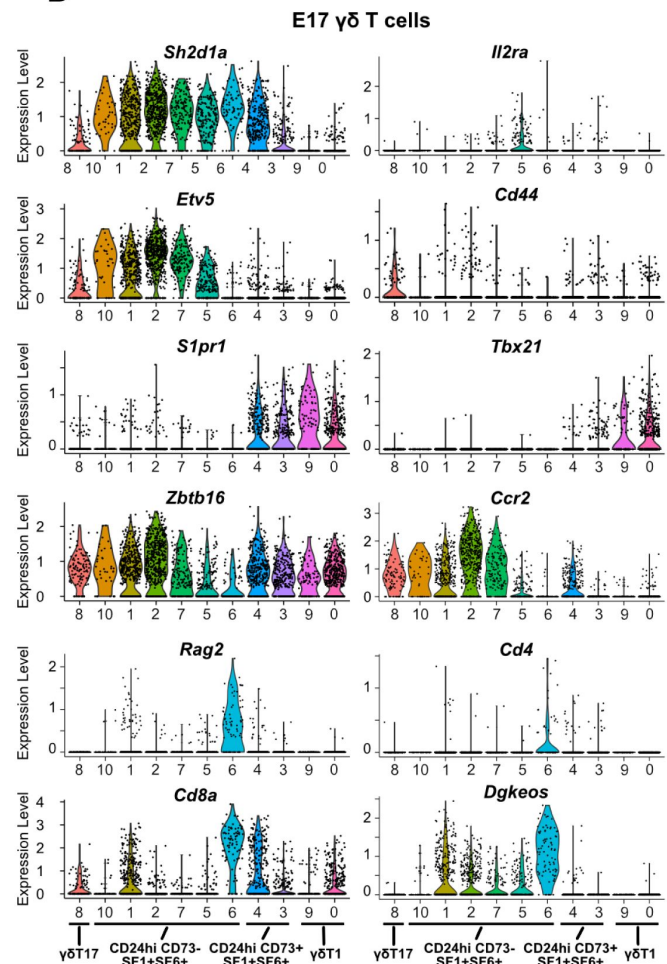
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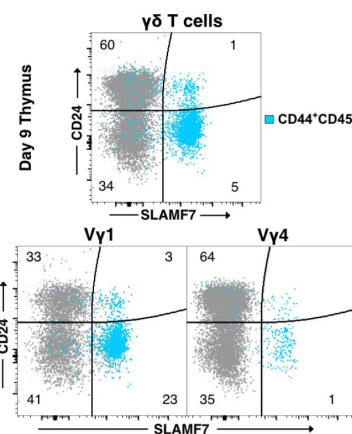
C



D



E



Supplemental Figure 1

Quality control for single cell CITE-seq.

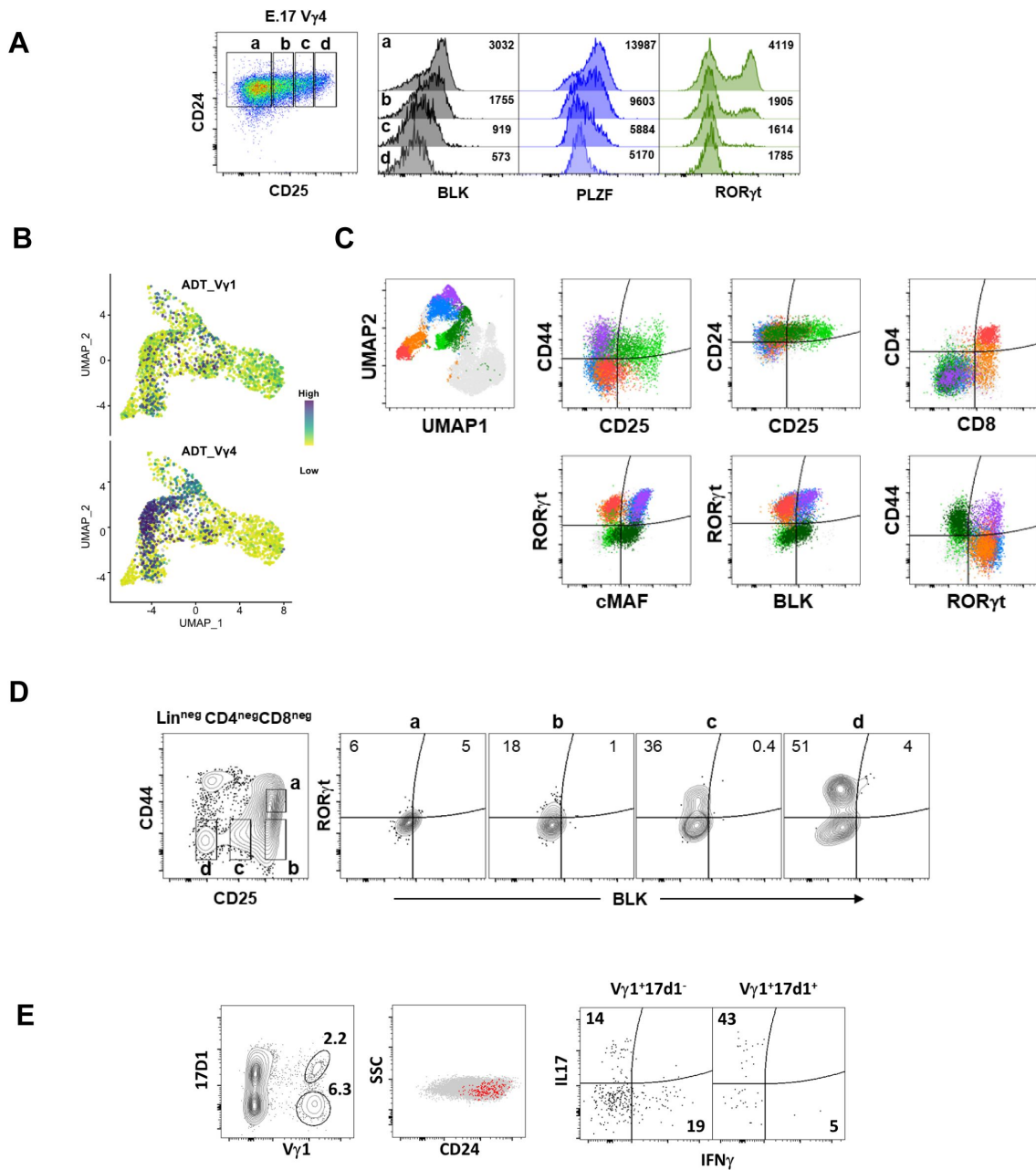
(A) Representative gating scheme used for FACS sorting $\gamma\delta$ cells from B6 (top) and B6.SAP^{-/-} (bottom) thymus. **(B)** *Left*, Violin plots of B6 E17 thymic $\gamma\delta$ T cells. nFeaure_RNA represents number of genes detected in each cell. nCount_RNA represents total number of molecules detected within a cell, percent.mt represents the frequency of mitochondrial genes expressed in each cell, and percent.ribo represents the frequency of ribosomal genes expressed in each cell. *Middle*, ridge plots demonstrating successful enrichment for selected hashtags in E17 thymus $\gamma\delta$ T cells following demultiplexing. *Right*, UMAP representations showing distribution of demultiplexed $\gamma\delta$ T cells among individual B6 E17 thymus samples. **(C)** Violin plots depicting distribution of cell surface protein oligo-conjugated antibody derived tags (ADTs) among different B6 E17 thymus $\gamma\delta$ T cells clusters. **(D)** Violin plots depicting expression levels of selected genes among individual E17 B6 thymic $\gamma\delta$ T cells clusters. **(E)** Representative dot plots of CD24 and SLAMF7 staining on neonatal thymus $\gamma\delta$ T cells. The position of CD44⁺CD45RB⁺ cells are shown in blue. Data are representative of two independent experiments, n = 10 mice.



Supplemental Figure 2

TCR repertoire profiling of B6 E17 thymus $\gamma\delta$ T cells.

(A) *Left*, TCR clonotype bubble plot depicting the top 100 TCR clonotypes among B6 E17 $\gamma\delta$ T cells ($n = 2502$ $\gamma\delta$ T cells). Each bubble represents a unique TRGV/TRDV clonotype; the bubble size corresponds to the clonotype frequency, and colors correspond to specific TRGV chains utilized. The 4 most frequent clonotypes are numbered and their corresponding CDR3 γ and CDR3 δ sequences are displayed below. *Right*, UMAP representations depicting the location of the four most frequent clonotypes E17 thymic $\gamma\delta$ T cell clusters. **(B)** Diversity of the E17 $\gamma\delta$ TCR repertoire, represented by the Inverse Simpson index. **(C)** Amino acid length distributions of E17 thymic $\gamma\delta$ T cell CDR3 γ (top) and CDR3 δ (bottom) in immature *Blk*⁺, *Etv5*⁺, *Maf*⁺ C2 and C7 (left), immature ROR γ t^{high} C1 (middle), and mature ROR γ t^{high} c8 (right) clusters. The top 10 clonotypes are color-coded, all other clonotypes are shown in gray.

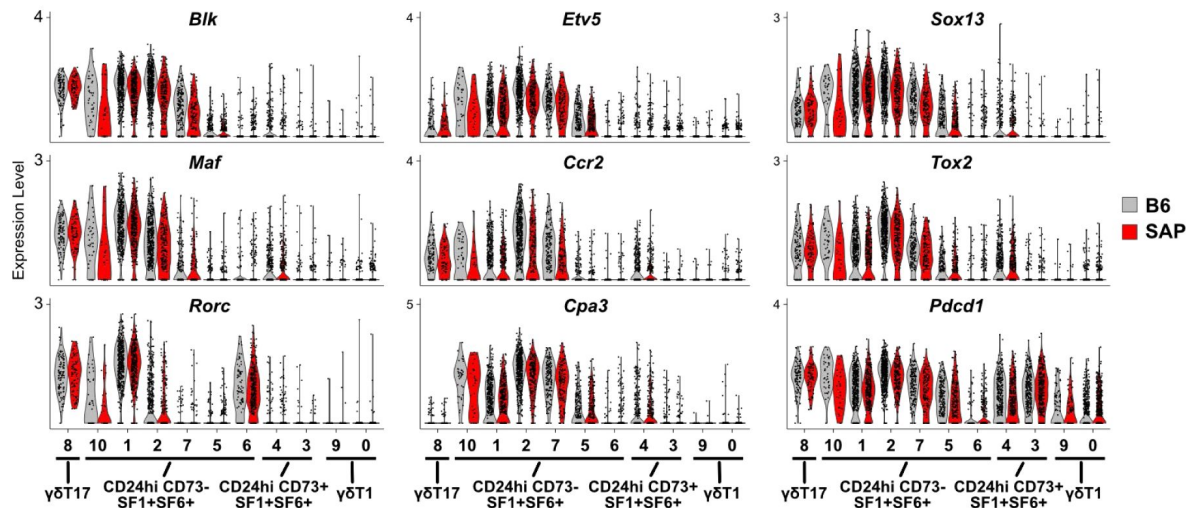


Supplemental Figure 3

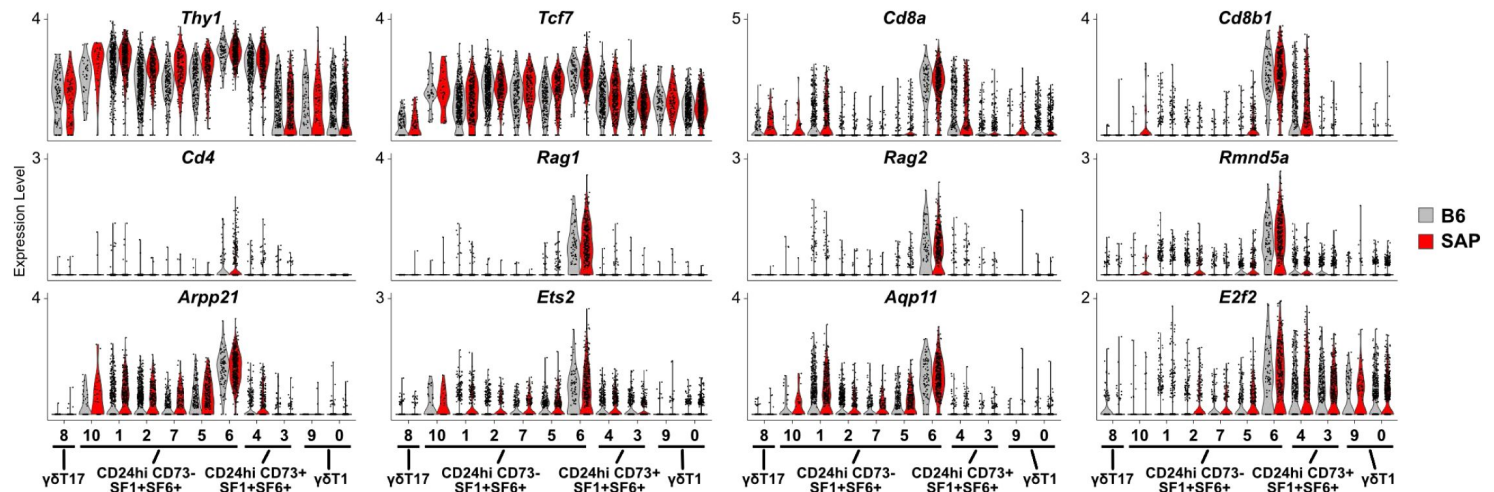
Identification of a BLK^{neg}MAF^{neg}RORγt^{pos} E17 γδ T cell population expressing CD4 and CD8.

(A) *Left*, Representative dot plot of CD24 and CD25 expression on E17 B6 thymic Vy4 T cells. *Right*, Histograms depicting BLK, PLZF, and RORγt expression in the gated populations at left. The geometric mean fluorescence intensity of individual populations is shown. **(B)** Feature plots showing cell surface protein expression of Vy1 (top) and Vy4 (bottom) on B6 E17 thymus γδ T cells. Each point represents a cell, color-coded based on the protein expression level (high: dark blue, low: yellow). **(C)** *Top left*, UMAP representation of E17 B6 γδ T cell flow cytometry data with selected FlowSOM clusters indicated by color. All other cells are shown in gray. *Right*, Individual dot plots overlaid with the selected FlowSOM clusters from the UMAP plot. Data represent concatenated data from 5 B6 TCRβ^{neg}TCRδ^{pos} γδ T cells and are representative of 2 independent experiments. **(D)** Increasing BLK^{neg}RORγt^{pos} frequency as thymocytes progress to DN4 stage. Representative contour plots of E17 DN thymocytes with selected gates shown (*left*) and BLK and RORγt expression associated with each gated population (*right*). Numbers indicate the percentage of cells in the gated population. **(E)** Immature Vy1Vδ1 T cells comprise the major IL17-producing Vy1 T cell population in E17 thymus. *Left*, representative contour plot of Vy1 and Vδ1 (17D1) expression among E17 thymic γδ T cells. *Middle*, Dot plot of CD24 and SSC expression on E17 thymic γδ T cells. The position of Vy1⁺Vδ1⁺ E17 T cells is shown in red. *Right*, Dot plots of IL-17 and IFN-γ expression in E17 thymic Vy1⁺17D1⁺ and Vy1⁺17D1⁻ cells after PMA/ionomycin stimulation.

A

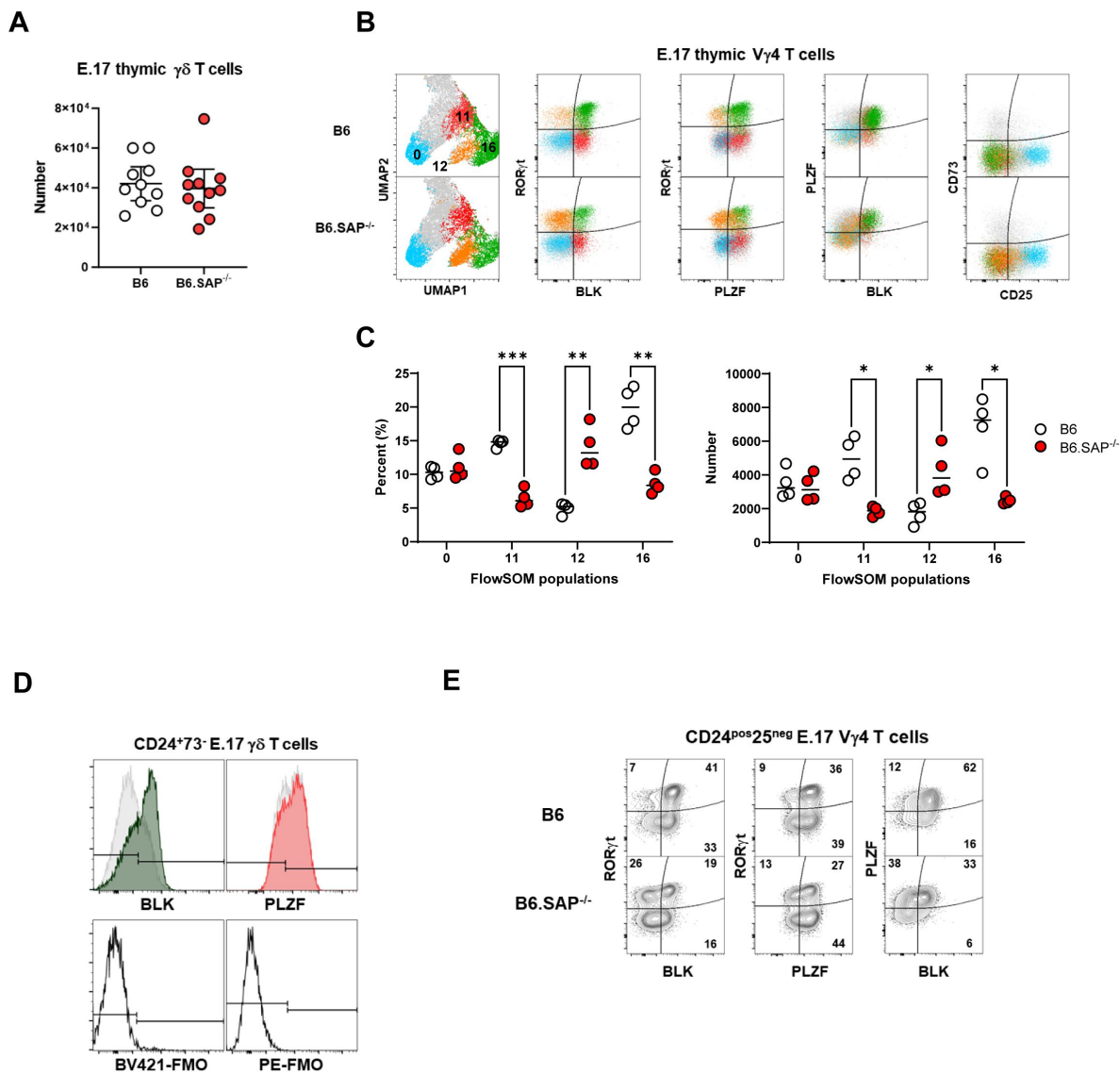


B



Supplemental Figure 4

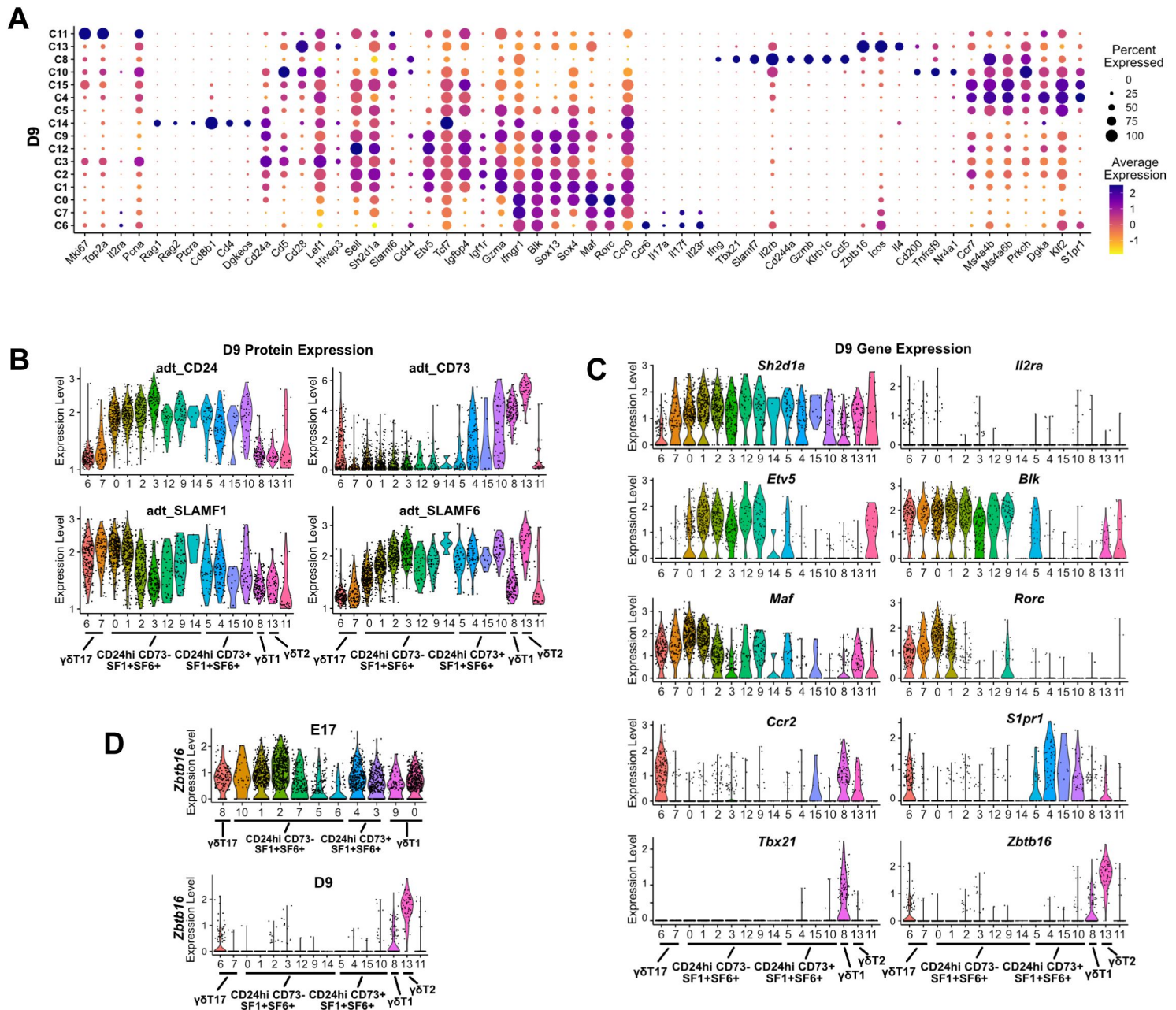
Violin plots showing expression levels of selected genes among individual B6 and B6.SAP^{-/-} E17 thymic γδ T cell clusters. **(A)** Genes whose expression decreased in B6.SAP^{-/-} immature γδ T cells. **(B)** Genes whose expression increased in B6.SAP^{-/-} immature γδ T cells.



Supplemental Figure 5

SAP-dependent regulation of immature BLK⁺PLZF⁺ROR γ t⁺ and BLK⁻PLZF⁺/ROR γ t⁺ E17 thymic $\gamma\delta$ T cells.

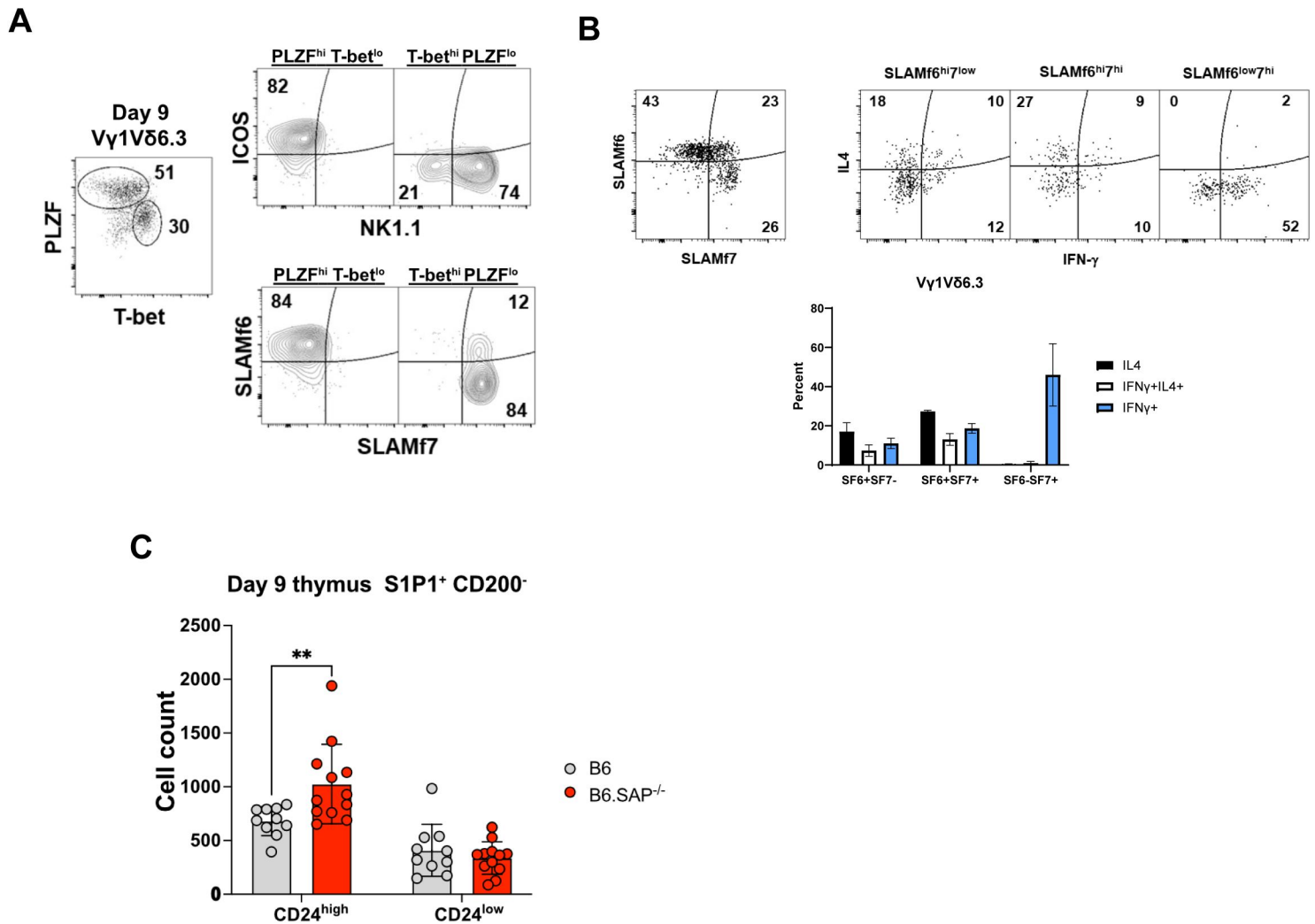
(A) Total number of E17 thymic $\gamma\delta$ T cells in B6 and B6.SAP^{-/-} E17 thymus. **(B)** E17 thymic V γ 4 developmental checkpoints. *Left*, UMAP representation of E17 B6 and B6.SAP^{-/-} V γ 4 T cell flow cytometry data with selected FlowSOM clusters indicated by color. All other cells are shown in gray. *Right*, Individual dot plots overlaid with the selected FlowSOM clusters from the UMAP plot. Data represent concatenated data from 4 B6 and 4 B6.SAP^{-/-} TCR β ^{neg}TCR δ ^{pos} $\gamma\delta$ T cells and are representative of 2 independent experiments. **(C)** Cumulative frequency and number of the indicated FlowSOM populations shown in (B), * $p \leq 0.05$, ** $p \leq 0.01$, *** $p \leq 0.001$, unpaired t-test, corrected for multiple comparison using Holm-Sidak's test. **(D)** Decreased BLK, but not PLZF expression in B6.SAP^{-/-} $\gamma\delta$ T cells. *Above*, histograms of BLK and PLZF in immature E17 $\gamma\delta$ T cells. B6 BLK (dark gray) and PLZF (red) histograms are overlaid on B6.SAP^{-/-} (light gray) histograms for comparison. *Below*, control stains lacking BLK or PLZF. **(E)** Representative contour plots of immature CD25^{neg} E.17 V γ 4 T cells in B6 and B6.SAP^{-/-} mice.



Supplemental Figure 6

Transcriptional heterogeneity in neonatal thymic $\gamma\delta$ T cells.

(A) Dot plot showing scaled expression level of selected genes in B6 D9 thymic $\gamma\delta$ T cells. Level of normalized expression is shown using a color scale ranging from low (yellow) to high (purple). Dot size represents the fraction of cells within each cluster that express the marker. (B) Violin plots showing the specificity of the oligo-conjugated antibody derived tags (ADT) to cell surface proteins (CD24, CD73, SLAMF1, SLAMF6) among different B6 D9 thymus $\gamma\delta$ T cell clusters. (C) Violin plots showing expression levels of selected genes among individual clusters in the B6 D9 thymic $\gamma\delta$ T cells. (D) Violin plots showing expression levels of *Zbtb16* gene (expressing PLZF) among individual clusters in the B6 E17 (top) and D9 (bottom) thymic $\gamma\delta$ T cells.

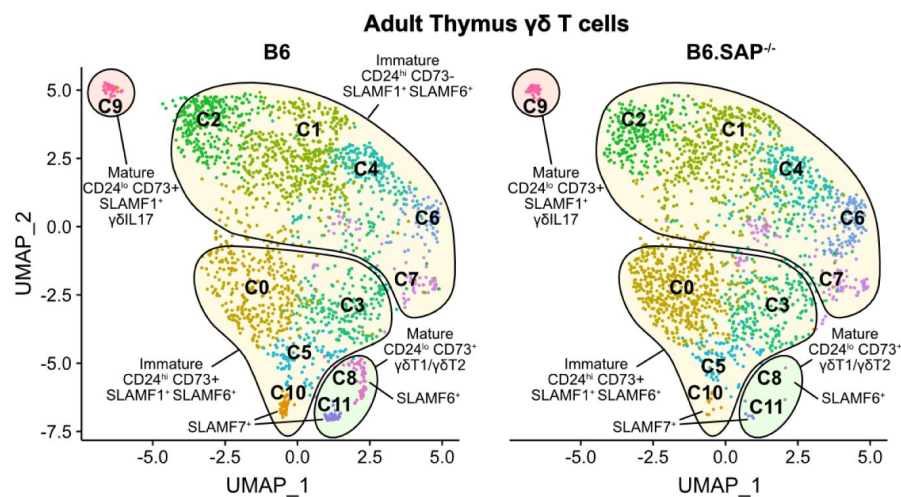


Supplemental Figure 7

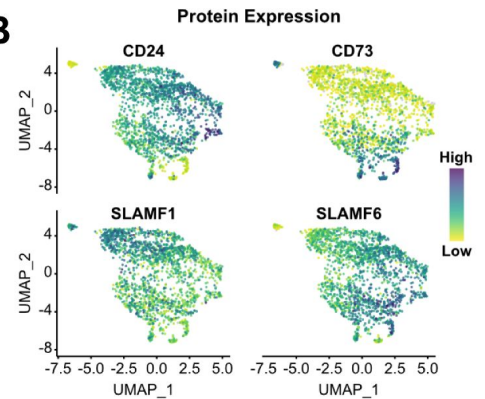
Flow cytometric analysis of D9 thymic $\gamma\delta$ T cells.

(A) SLAMF6 and SLAMF7 expression defines $\gamma\delta$ NKT cell subsets. *Left*, representative dot plot of PLZF and T-bet expression in B6 D9 thymic V γ 1V δ 6.3 $\gamma\delta$ NKT cells. Representative contour plots of ICOS and NK1.1 (*top right*) and SLAMF6 and SLAMF7 (*bottom right*) staining in PLZF^{hi}T-bet^{lo} and PLZF^{lo}T-bet^{hi} subsets are shown at right. Data are representative of two independent experiments with a total of ten mice. **(B)** *Left*, representative dot plot of SLAMF6 and SLAMF7 expression in B6 D9 thymic V γ 1V δ 6.3 $\gamma\delta$ NKT cells. *Right*, representative dot plots of intracellular IL4 and IFN- γ expression in SLAMF6^{hi}SLAMF7^{low}, SLAMF6^{hi}SLAMF7^{hi}, and SLAMF6^{low}SLAMF7^{hi} $\gamma\delta$ NKT cell subsets. Numbers indicate percentage. Cumulative data of IL-4 and IFN- γ expression is shown below. Data are representative of two independent experiments with a total of ten mice. **(C)** Increased number of immature CD200⁻S1P1⁺ $\gamma\delta$ T cells in SAP-deficient neonatal thymus.

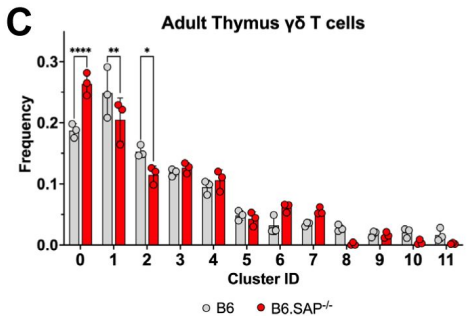
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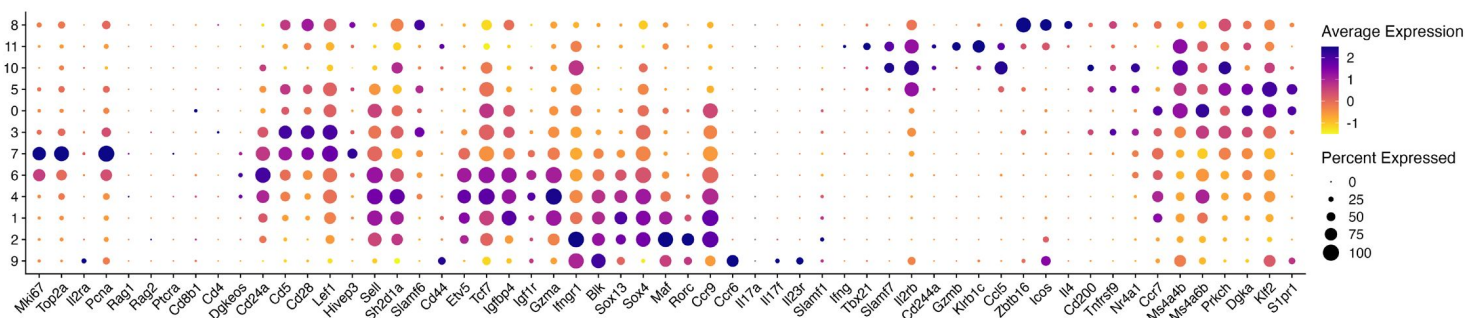
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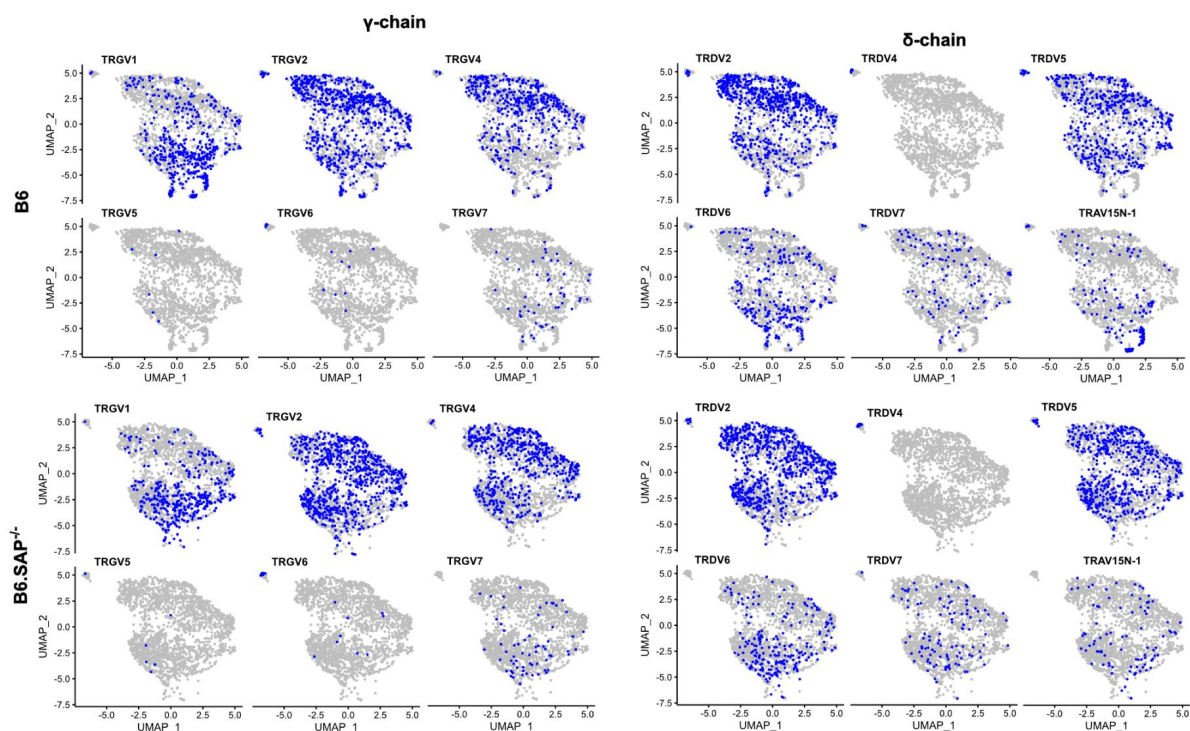
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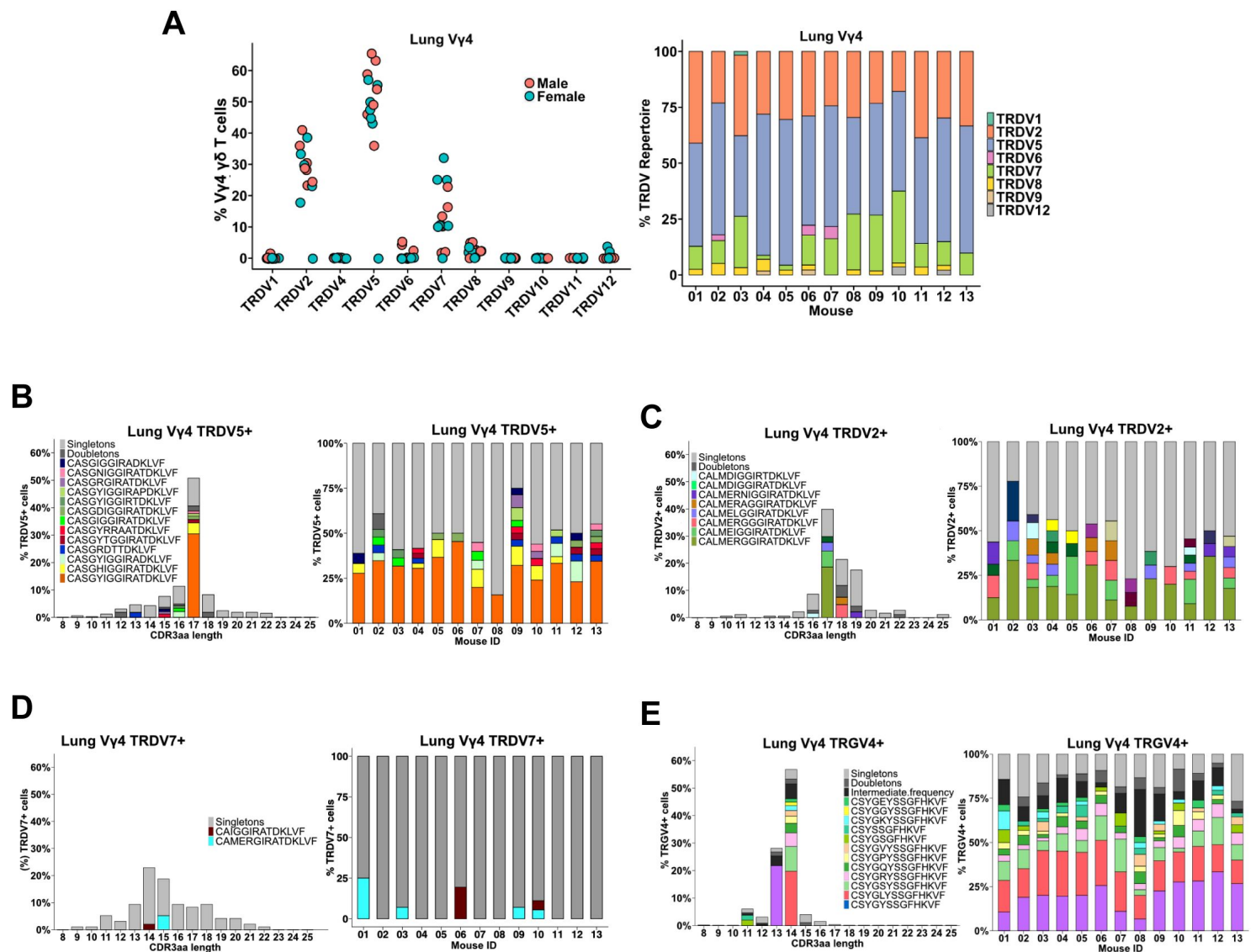
E



Supplemental Figure 8

scRNA-seq with TCR repertoire profiling of B6 and B6.SAP^{-/-} adult thymic γδ T cells.

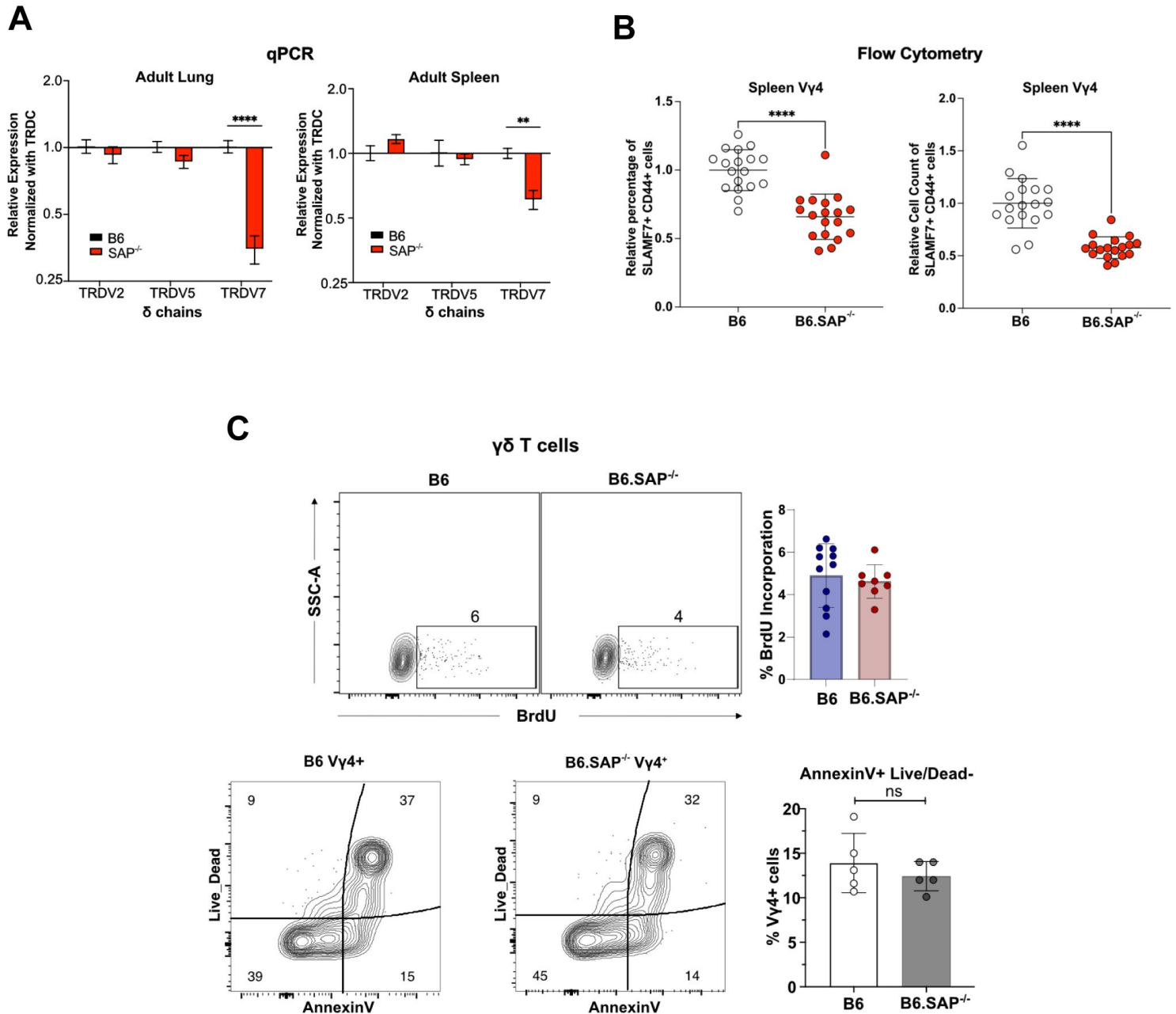
(A) UMAP representation of B6 and B6.SAP^{-/-} adult thymus (6 weeks old) γδ T cells, n = 3 mice per strain. **(B)** Feature plots illustrating the cell surface protein expression profiles of CD24, CD73, SLAMF1, and SLAMF6 on adult thymic γδ T cells. Each data point represents a cell, color-coded to indicate varying protein expression levels (high: dark blue, low: yellow). **(C)** Frequencies of B6 and B6.SAP^{-/-} γδ T cell clusters. Bars represent the mean cluster frequency, error bars represent standard deviation, *p ≤ 0.05, **p ≤ 0.01, ****p ≤ 0.0001, two-way ANOVA, Sidak multiple-comparisons test. **(D)** Dot plot showing scaled expression level of selected genes in B6 adult thymic γδ T cells. Level of normalized expression is shown using a color scale ranging from low (yellow) to high (purple). Dot size represents the fraction of cells within each cluster that express the marker. **(E)** Expression of selected TRG (left) and TRDs chain (right) V-segment usage (blue) by individual adult thymus γδ T cells in B6 and B6.SAP^{-/-} mice.



Supplemental Figure 9

Lung Vy4 exhibit a high usage of TRDV7 chains with a diverse CDR3.

(A) TRDV usage among FACS-sorted lung Vy4 $\gamma\delta$ T cells from 7 male and 6 female mice. (B) *Left*, amino acid length distribution of top lung TRDV5⁺ CDR3 δ clonotypes in the lung Vy4 $\gamma\delta$ T cell population. *Right*, distribution of the most frequent clonotypes among individual B6 mice. (C) *Left*, amino acid length distribution of top lung TRDV2⁺ CDR3 δ clonotypes in the lung Vy4 $\gamma\delta$ T cell population. *Right*, distribution of the most frequent clonotypes among individual B6 mice. (D) *Left*, amino acid length distribution of top lung TRDV7⁺ CDR3 δ clonotypes in the lung Vy4 $\gamma\delta$ T cell population. *Right*, distribution of the most frequent clonotypes among individual B6 mice. (E) *Left*, amino acid length distribution of top lung TRGV4⁺ CDR3 γ clonotypes in the lung Vy4 $\gamma\delta$ T cell population. *Right*, distribution of the most frequent clonotypes among individual B6 mice.



Supplemental Figure 10

Significant and specific reduction in peripheral TRGV4/TRDV7⁺ $\gamma\delta$ T1 in $B6.SAP^{-/-}$ mice.

(A) The relative expression of the specified TRDV transcripts in adult lung and spleen CD45⁺ leukocytes (normalized to total TRDC transcript levels) in B6 and $B6.SAP^{-/-}$ mice, ** $p < 0.01$, **** $p < 0.0001$, two-way ANOVA followed by Sidak multiple-comparisons test. (B) SAP-dependent decrease in CD44⁺SLAMF7⁺ spleen Vy4 T cells. The relative percentage and count of CD44⁺SLAMF7⁺ Vy4 cells in B6 and $B6.SAP^{-/-}$ spleen subsets are shown. Data are representative of five independent experiments with 3-5 mice per strain per experiment, **** $p, < 0.0001$, as determined by unpaired t tests. (C) No difference in lung Vy4 homeostasis in $B6.SAP^{-/-}$ mice. *Top left*, Representative contour plots showing the frequency of BrdU staining in B6 and $B6.SAP^{-/-}$ lung $\gamma\delta$ T cells. Cumulative data from two independent experiments are shown at *top right*. *Bottom left*, representative contour plots depicting Annexin V and Live/Dead staining in B6 and $B6.SAP^{-/-}$ lung $\gamma\delta$ T cells. Cumulative results are shown at *bottom right*, and are representative of 3 independent experiments.

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Article and author information

Somen K Mistri

Department of Surgery, Larner College of Medicine, University of Vermont, Burlington, Vermont 05405, USA
ORCID iD: [0000-0002-2139-6927](https://orcid.org/0000-0002-2139-6927)

Brianna M. Hilton

Department of Surgery, Larner College of Medicine, University of Vermont, Burlington, Vermont 05405, USA

Katherine J. Horrigan

Department of Surgery, Larner College of Medicine, University of Vermont, Burlington, Vermont 05405, USA

Emma S. Andretta

Department of Surgery, Larner College of Medicine, University of Vermont, Burlington, Vermont 05405, USA
ORCID iD: [0009-0006-6827-1933](https://orcid.org/0009-0006-6827-1933)

Remi Savard

Department of Surgery, Larner College of Medicine, University of Vermont, Burlington, Vermont 05405, USA

Oliver Dienz

Department of Surgery, Larner College of Medicine, University of Vermont, Burlington, Vermont 05405, USA
ORCID iD: [0000-0001-9380-4873](https://orcid.org/0000-0001-9380-4873)

Kenneth J Hampel

Department of Pathology and Laboratory Medicine, Larner College of Medicine, University of Vermont Medical Center, Burlington, Vermont 05405, USA

Diana L. Gerrard

Department of Pathology and Laboratory Medicine, Larner College of Medicine, University of Vermont Medical Center, Burlington, Vermont 05405, USA

Joshua T. Rose

Department of Pathology and Laboratory Medicine, Larner College of Medicine, University of Vermont Medical Center, Burlington, Vermont 05405, USA

Nikoletta Sidiropoulos

Department of Pathology and Laboratory Medicine, Larner College of Medicine, University of Vermont Medical Center, Burlington, Vermont 05405, USA

Devdoot Majumdar

Department of Surgery, Larner College of Medicine, University of Vermont, Burlington, Vermont 05405, USA

ORCID iD: [0000-0002-2541-3851](https://orcid.org/0000-0002-2541-3851)

Jonathan E. Boyson

Department of Surgery, Larner College of Medicine, University of Vermont, Burlington, Vermont 05405, USA

For correspondence: jboyson@uvm.edu

ORCID iD: [0000-0003-2673-9148](https://orcid.org/0000-0003-2673-9148)

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Reviewer #1 (Public Review):

Summary:

In this study, the authors advance their previous findings on the role of the SLAM-SAP signaling pathway in the development and function of multiple innate-like gamma-delta T-cell subsets. Using a high throughput single-cell proteogenomics approach, the authors uncover SAP-dependent developmental checkpoints, and the role of SAP signaling in regulating the diversion of $\gamma\delta$ T cells into the $\alpha\beta$ T cell developmental pathway. Finally, the authors define TRGV4/TRAV13-4(DV7)-expressing T cells as a novel, SAP-dependent V γ 4 $\gamma\delta$ T1 subset.

Strengths:

This study furthers our understanding of the importance and complexity of the SLAM-SAP signaling pathway not only in the development of innate-like $\gamma\delta$ T cells but also in how it potentially balances the $\gamma\delta/\alpha\beta$ T cell lineage commitment. Additionally, this study reveals the role of SAP-dependent events in the generation of $\gamma\delta$ TCR repertoire.

The conclusions of the study are supported by well-thought-out experiments and compelling data.

Weaknesses:

No major weaknesses in the study were identified.

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Reviewer #2 (Public Review):

Summary:

Mistri et al explore the role of SLAM-SAP signaling in the developmental programming of innate-like gd T cell subsets. Using proteo-genomics, they determined that abrogation of SLAM-SAP signaling altered that programming, reducing some IL-17-producing subsets, including a novel V γ 4 $\gamma\delta$ T1 subset, and diverting gdTCR-expressing precursors to the ab fate. Altogether, this is a very thorough, thoughtfully interpreted study that adds significantly to our understanding of the contribution of the SLAM-SAP pathway to lineage specification. A particularly interesting element is the role of SLAM-SAP in preventing gd17 progenitors from switching fates and adopting the ab fate.

One thing to keep in mind in assessing the ultimate fate of the "ab wannabe cells" is that mechanisms exist to silence the gd TCR as cells differentiate to the DP stage and so their presence as diverted DP cells may not be evident by staining for gdTCR expression - and will only be evident transcriptomically.

Strengths:

This is an exceedingly well-designed and thorough study that significantly enriches our understanding of gd T cell development.

Weaknesses:

There are no major weaknesses identified by this reviewer.

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